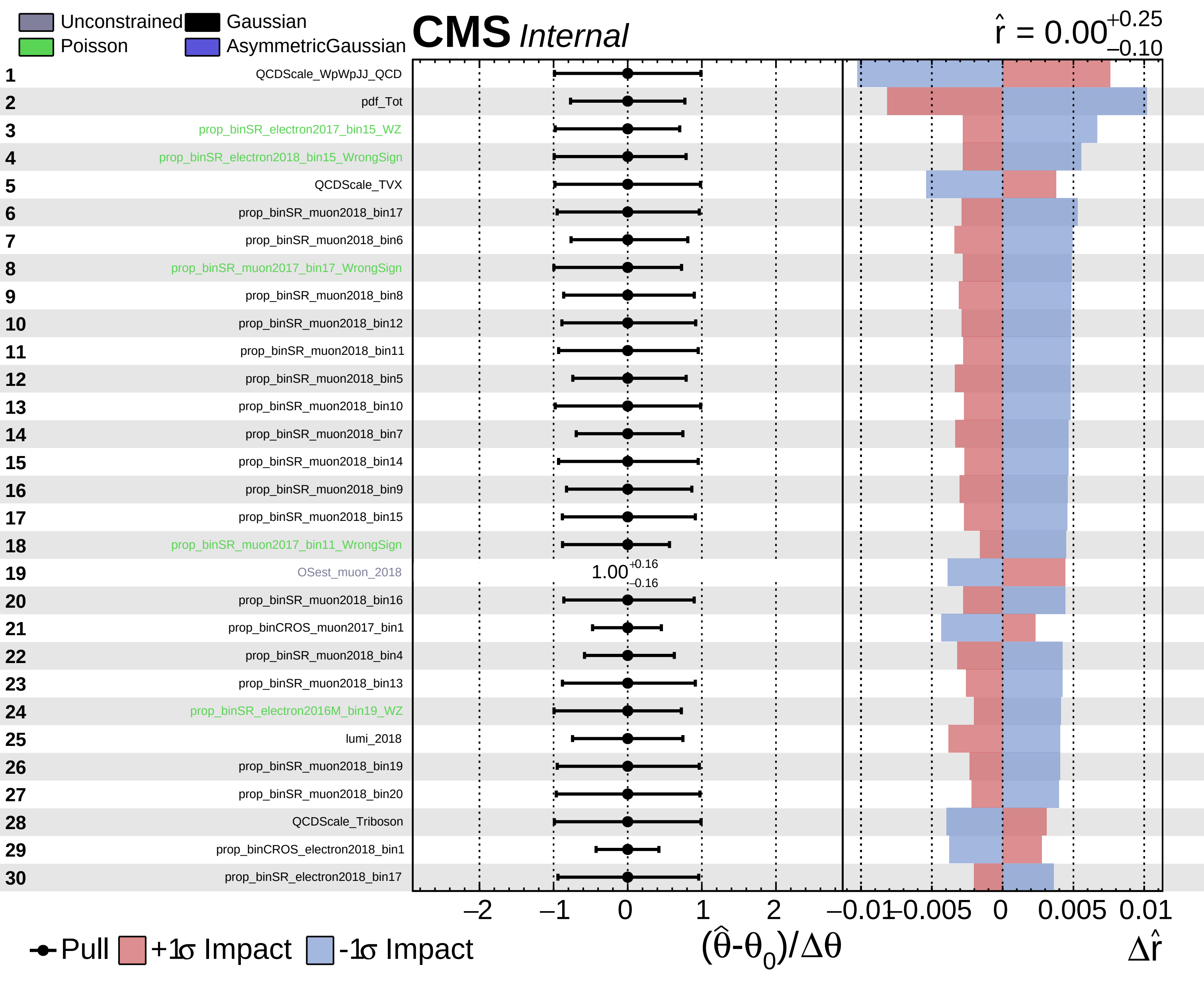
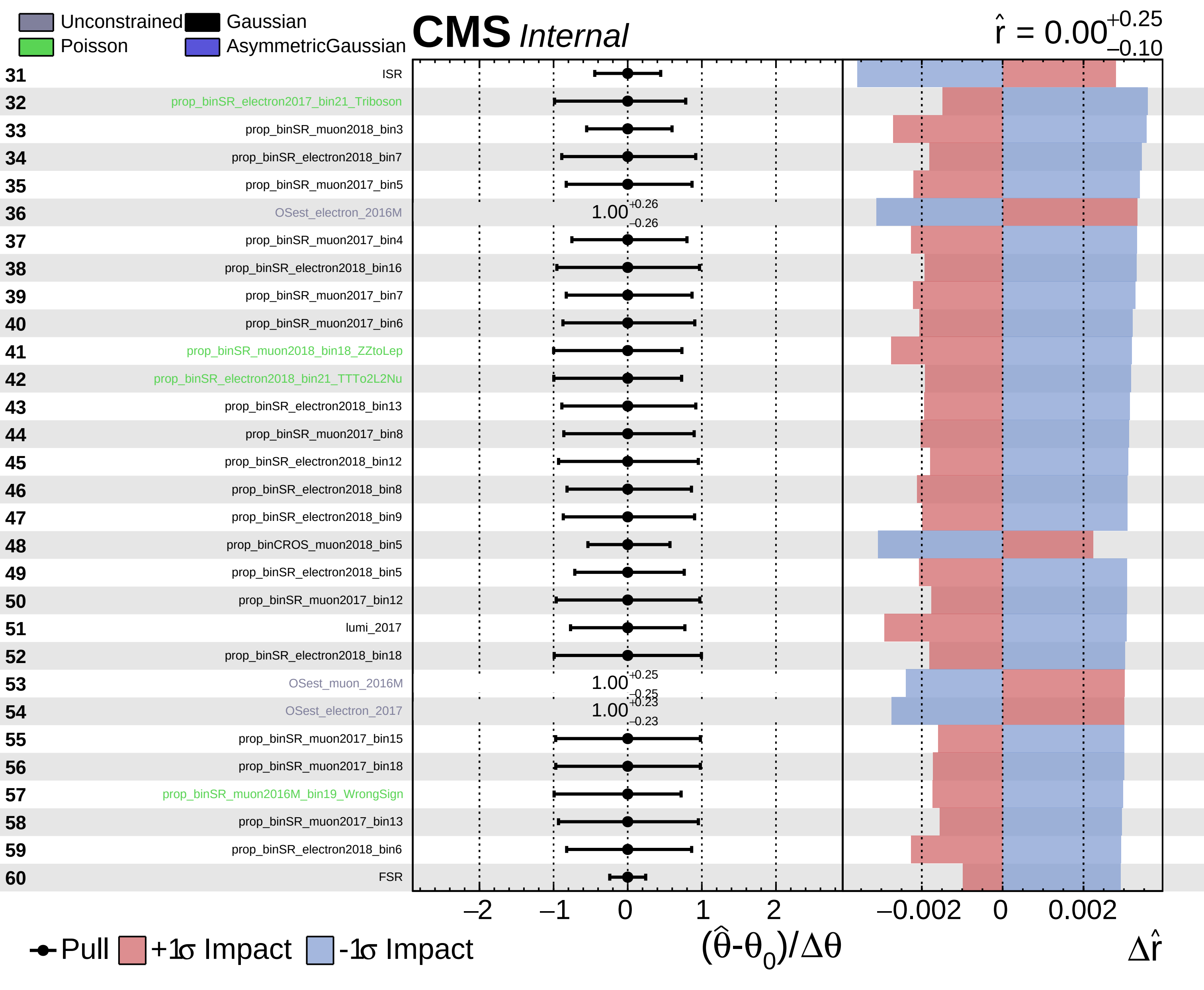
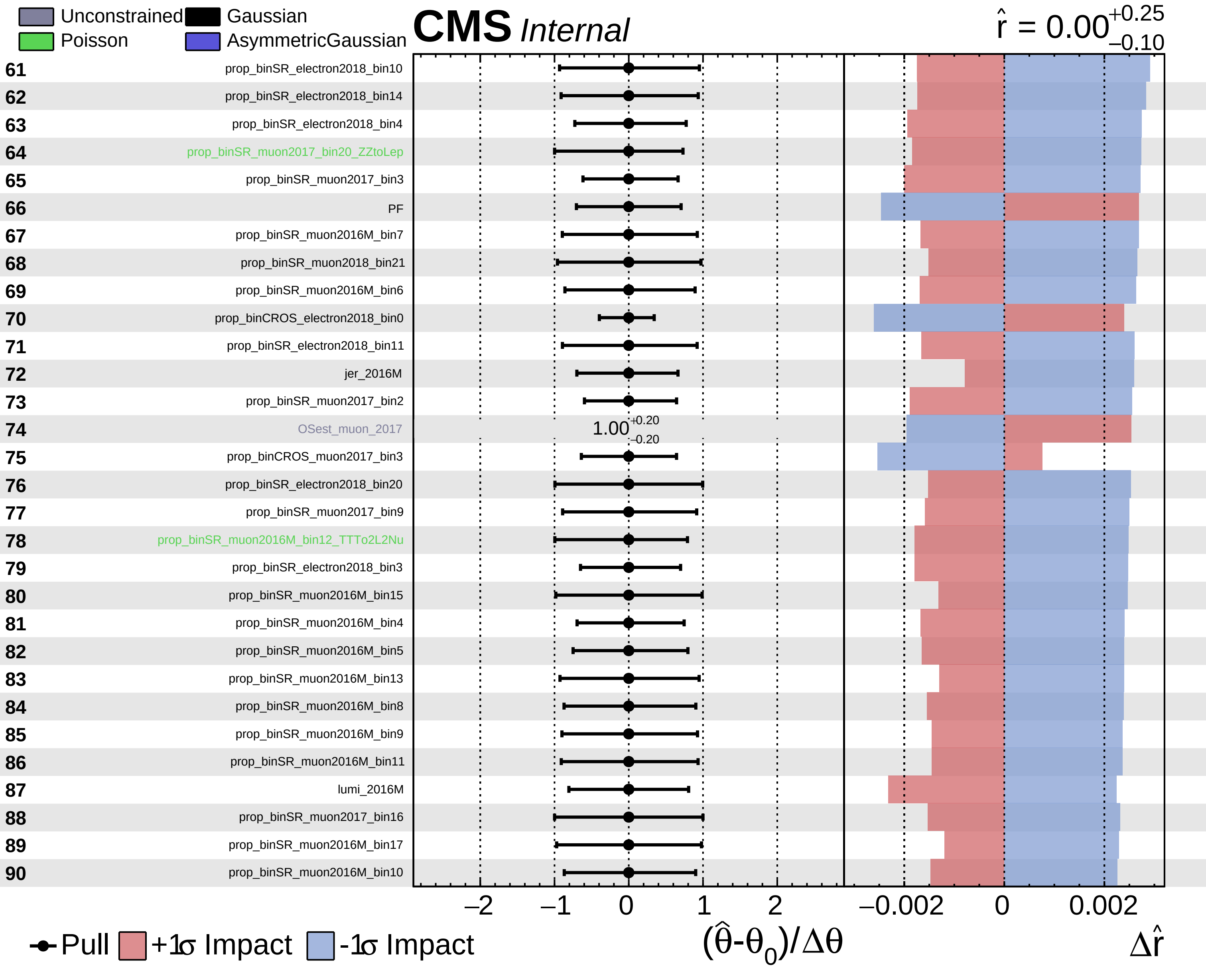


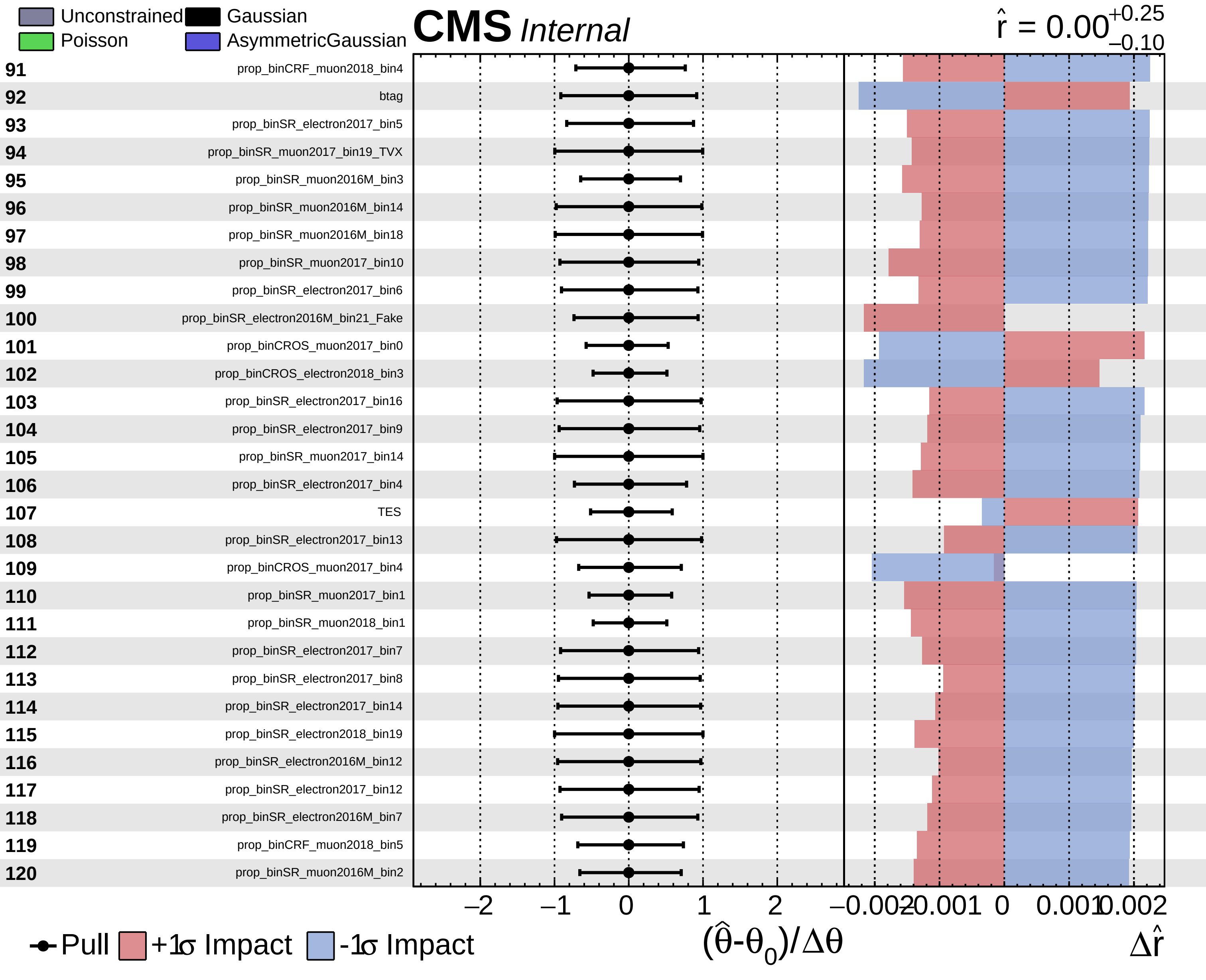
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.10
1 QCDScale_WpWpJJ_QCD
2 pdf_Tot
3 prop_binSR_electron2017_bin15_WZ
4 prop_binSR_electron2018_bin15_WrongSign
5 QCDScale_TVX
6 prop_binSR_muon2018_bin17
7 prop_binSR_muon2018_bin6
8 prop_binSR_muon2017_bin17_WrongSign
9 prop_binSR_muon2018_bin8
10 prop_binSR_muon2018_bin12
11 prop_binSR_muon2018_bin11
12 prop_binSR_muon2018_bin5
13 prop_binSR_muon2018_bin10
14 prop_binSR_muon2018_bin7
15 prop_binSR_muon2018_bin14
16 prop_binSR_muon2018_bin9
17 prop_binSR_muon2018_bin15
18 prop_binSR_muon2017_bin11_WrongSign
19 OSest_muon_2018
1.00 +0.16 −0.16
20 prop_binSR_muon2018_bin16
21 prop_binCROS_muon2017_bin1
22 prop_binSR_muon2018_bin4
23 prop_binSR_muon2018_bin13
24 prop_binSR_electron2016M_bin19_WZ
25 lumi_2018
26 prop_binSR_muon2018_bin19
27 prop_binSR_muon2018_bin20
28 QCDScale_Triboson
29 prop_binCROS_electron2018_bin1
30 prop_binSR_electron2018_bin17
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



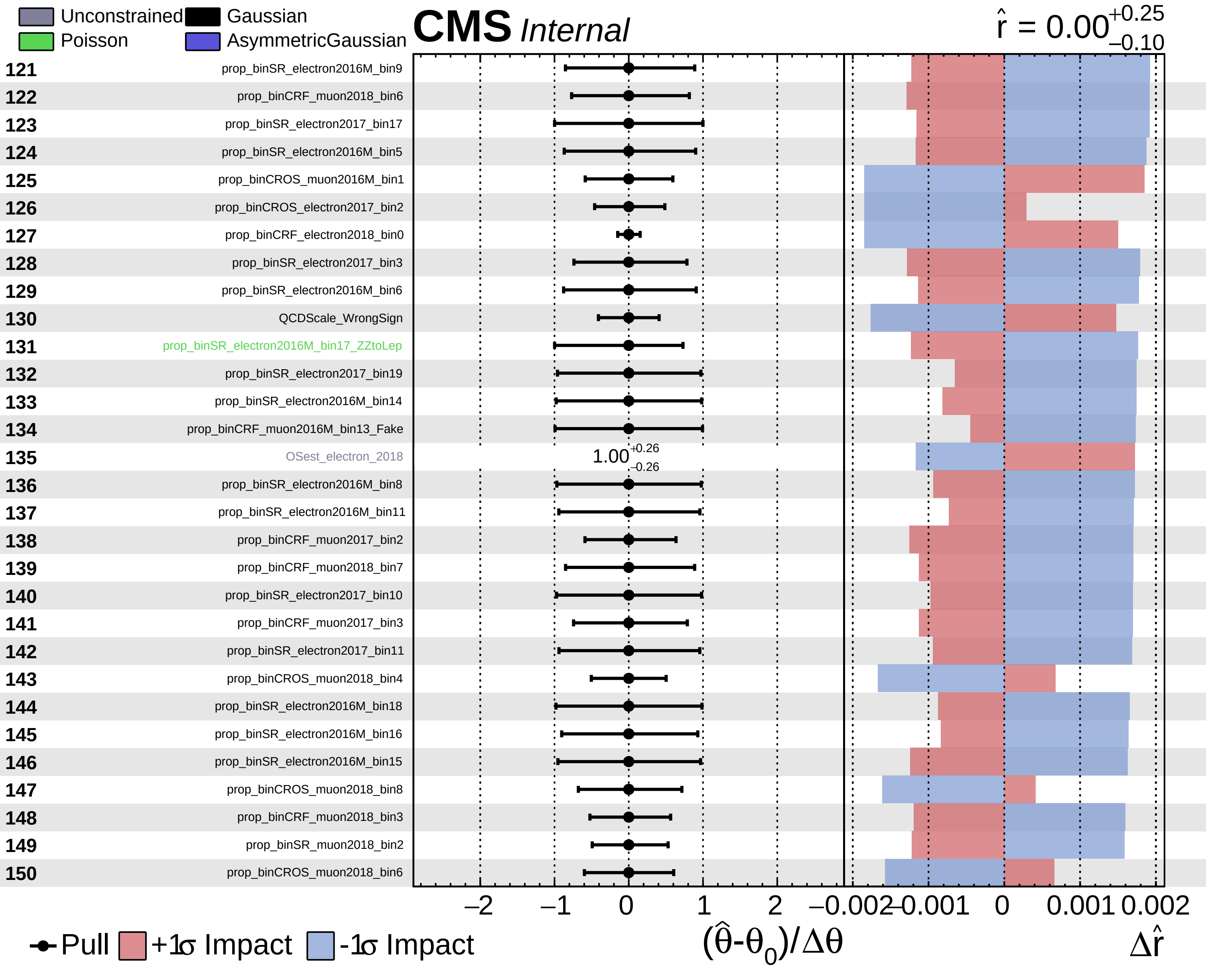
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
31 ISR
32 prop_binSR_electron2017_bin21_Triboson
33 prop_binSR_muon2018_bin3
34 prop_binSR_electron2018_bin7
35 prop_binSR_muon2017_bin5
36 OSest_electron_2016M
$1.00^{+0.26}_{-0.26}$
37 prop_binSR_muon2017_bin4
38 prop_binSR_electron2018_bin16
39 prop_binSR_muon2017_bin7
40 prop_binSR_muon2017_bin6
41 prop_binSR_muon2018_bin18_ZZtoLep
42 prop_binSR_electron2018_bin21_TTTo2L2Nu
43 prop_binSR_electron2018_bin13
44 prop_binSR_muon2017_bin8
45 prop_binSR_electron2018_bin12
46 prop_binSR_electron2018_bin8
47 prop_binSR_electron2018_bin9
48 prop_binCROS_muon2018_bin5
49 prop_binSR_electron2018_bin5
50 prop_binSR_muon2017_bin12
51 lumi_2017
52 prop_binSR_electron2018_bin18
53 OSest_muon_2016M
$1.00^{+0.25}_{-0.25}$
54 OSest_electron_2017
$1.00^{+0.23}_{-0.23}$
55 prop_binSR_muon2017_bin15
56 prop_binSR_muon2017_bin18
57 prop_binSR_muon2016M_bin19_WrongSign
58 prop_binSR_muon2017_bin13
59 prop_binSR_electron2018_bin6
60 FSR
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



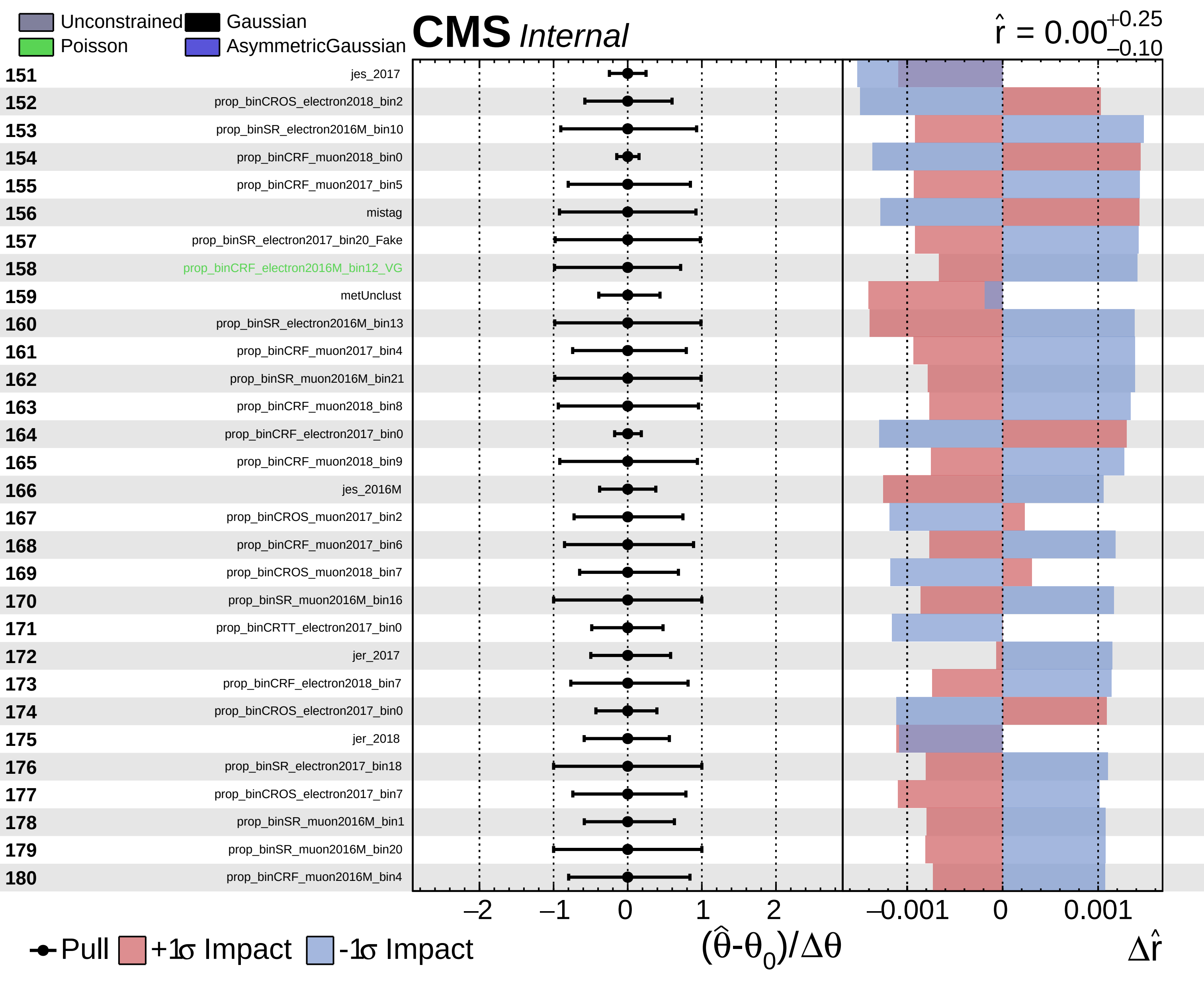
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
61 prop_binSR_electron2018_bin10
62 prop_binSR_electron2018_bin14
63 prop_binSR_electron2018_bin4
64 prop_binSR_muon2017_bin20_ZZtoLep
65 prop_binSR_muon2017_bin3
66 PF
67 prop_binSR_muon2016M_bin7
68 prop_binSR_muon2018_bin21
69 prop_binSR_muon2016M_bin6
70 prop_binCROS_electron2018_bin0
71 prop_binSR_electron2018_bin11
72 jer_2016M
73 prop_binSR_muon2017_bin2
74 OSest_muon_2017
$1.00^{+0.20}_{-0.20}$
75 prop_binCROS_muon2017_bin3
76 prop_binSR_electron2018_bin20
77 prop_binSR_muon2017_bin9
78 prop_binSR_muon2016M_bin12_TTTo2L2Nu
79 prop_binSR_electron2018_bin3
80 prop_binSR_muon2016M_bin15
81 prop_binSR_muon2016M_bin4
82 prop_binSR_muon2016M_bin5
83 prop_binSR_muon2016M_bin13
84 prop_binSR_muon2016M_bin8
85 prop_binSR_muon2016M_bin9
86 prop_binSR_muon2016M_bin11
87 lumi_2016M
88 prop_binSR_muon2017_bin16
89 prop_binSR_muon2016M_bin17
90 prop_binSR_muon2016M_bin10
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

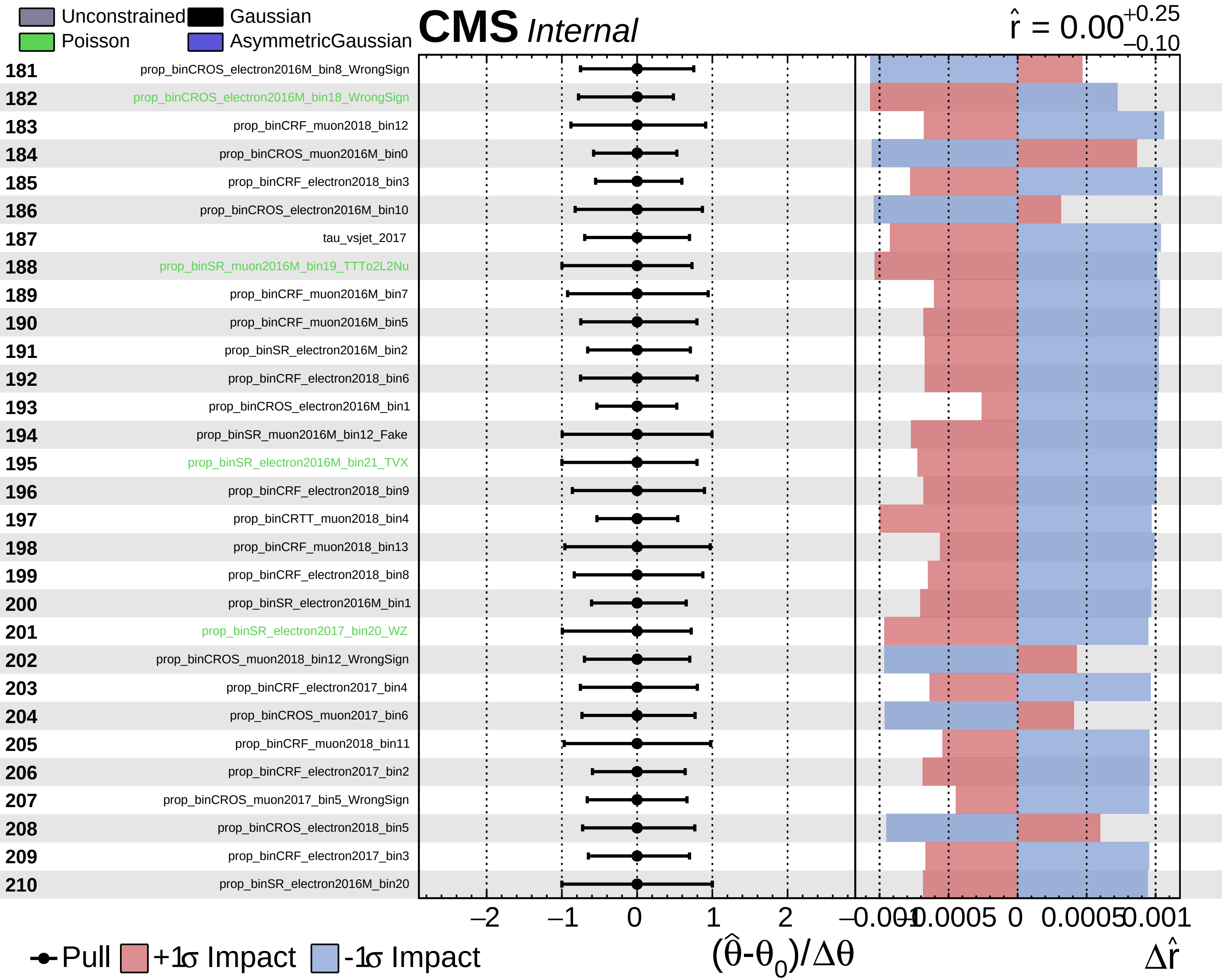


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
91 prop_binCRF_muon2018_bin4
92 btag
93 prop_binSR_electron2017_bin5
94 prop_binSR_muon2017_bin19_TVX
95 prop_binSR_muon2016M_bin3
96 prop_binSR_muon2016M_bin14
97 prop_binSR_muon2016M_bin18
98 prop_binSR_muon2017_bin10
99 prop_binSR_electron2017_bin6
100 prop_binSR_electron2016M_bin21_Fake
101 prop_binCROS_muon2017_bin0
102 prop_binCROS_electron2018_bin3
103 prop_binSR_electron2017_bin16
104 prop_binSR_electron2017_bin9
105 prop_binSR_muon2017_bin14
106 prop_binSR_electron2017_bin4
107 TES
108 prop_binSR_electron2017_bin13
109 prop_binCROS_muon2017_bin4
110 prop_binSR_muon2017_bin1
111 prop_binSR_muon2018_bin1
112 prop_binSR_electron2017_bin7
113 prop_binSR_electron2017_bin8
114 prop_binSR_electron2017_bin14
115 prop_binSR_electron2018_bin19
116 prop_binSR_electron2016M_bin12
117 prop_binSR_electron2017_bin12
118 prop_binSR_electron2016M_bin7
119 prop_binCRF_muon2018_bin5
120 prop_binSR_muon2016M_bin2
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

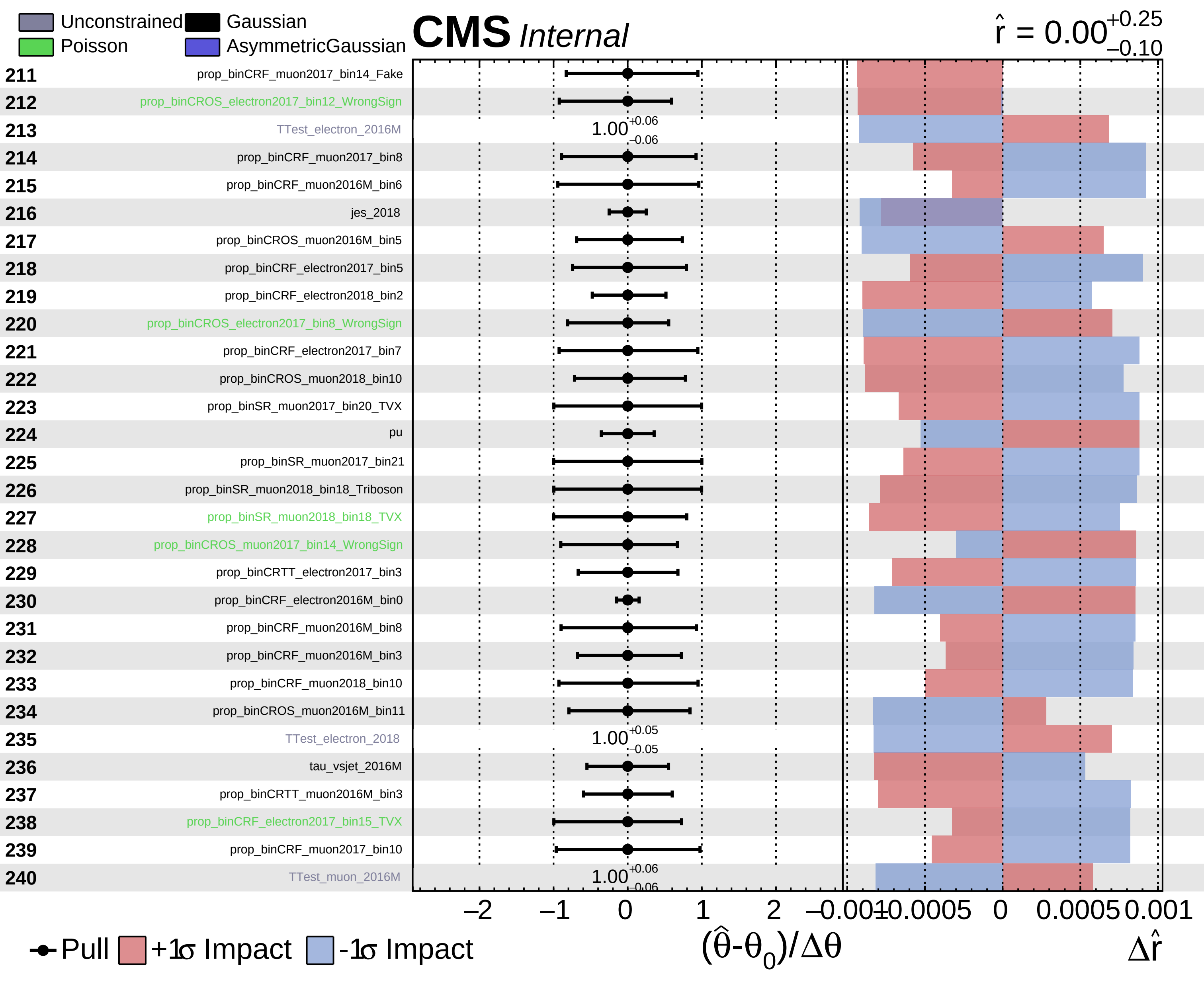


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
121 prop_binSR_electron2016M_bin9
122 prop_binCRF_muon2018_bin6
123 prop_binSR_electron2017_bin17
124 prop_binSR_electron2016M_bin5
125 prop_binCROS_muon2016M_bin1
126 prop_binCROS_electron2017_bin2
127 prop_binCRF_electron2018_bin0
128 prop_binSR_electron2017_bin3
129 prop_binSR_electron2016M_bin6
130 QCDScale_WrongSign
131 prop_binSR_electron2016M_bin17_ZZtoLep
132 prop_binSR_electron2017_bin19
133 prop_binSR_electron2016M_bin14
134 prop_binCRF_muon2016M_bin13_Fake
135 OSest_electron_2018
$1.00^{+0.26}_{-0.26}$
136 prop_binSR_electron2016M_bin8
137 prop_binSR_electron2016M_bin11
138 prop_binCRF_muon2017_bin2
139 prop_binCRF_muon2018_bin7
140 prop_binSR_electron2017_bin10
141 prop_binCRF_muon2017_bin3
142 prop_binSR_electron2017_bin11
143 prop_binCROS_muon2018_bin4
144 prop_binSR_electron2016M_bin18
145 prop_binSR_electron2016M_bin16
146 prop_binSR_electron2016M_bin15
147 prop_binCROS_muon2018_bin8
148 prop_binCRF_muon2018_bin3
149 prop_binSR_muon2018_bin2
150 prop_binCROS_muon2018_bin6
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

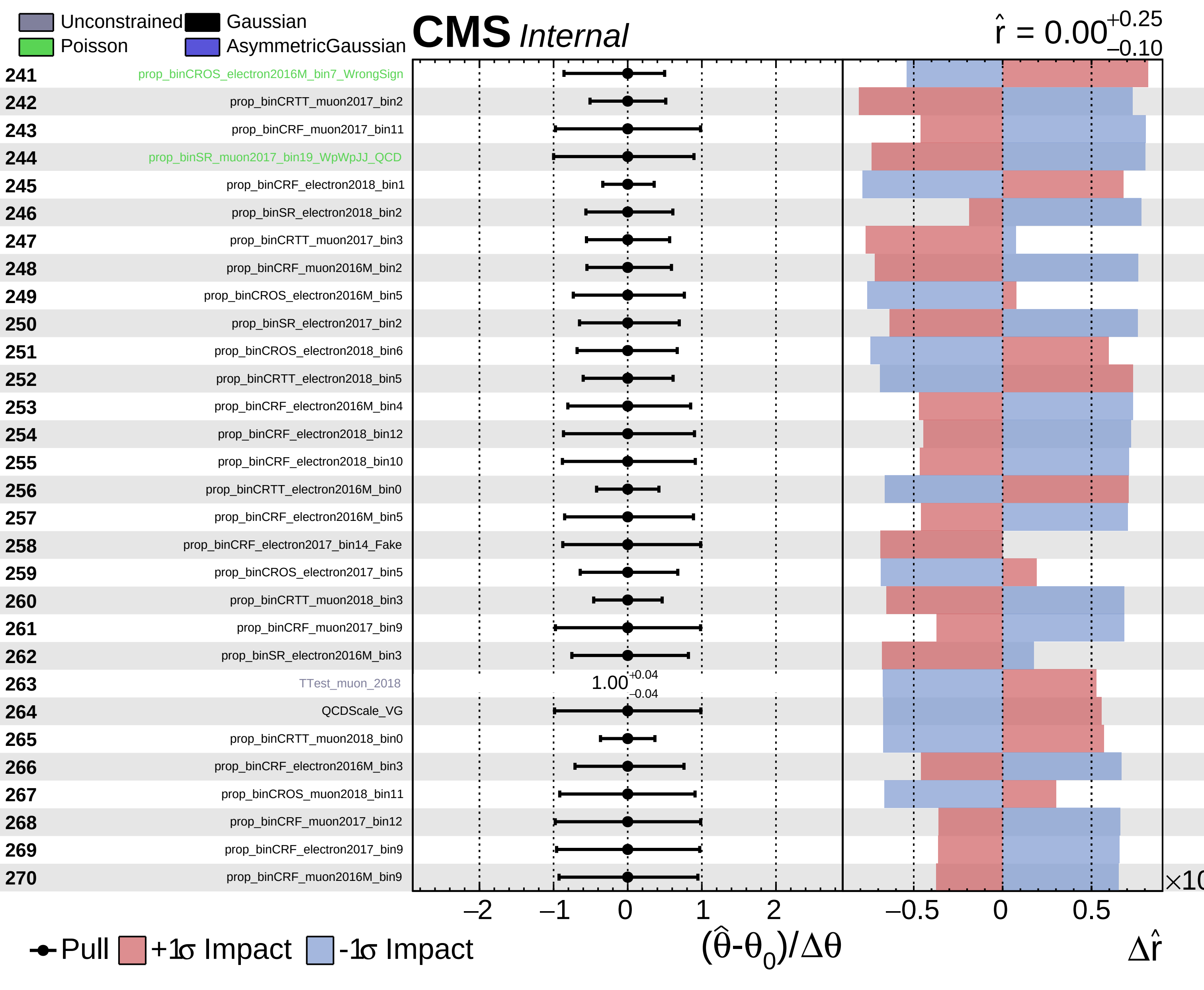

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
151 jes_2017
152 prop_binCROS_electron2018_bin2
153 prop_binSR_electron2016M_bin10
154 prop_binCRF_muon2018_bin0
155 prop_binCRF_muon2017_bin5
156 mistag
157 prop_binSR_electron2017_bin20_Fake
158 prop_binCRF_electron2016M_bin12_VG
159 metUnclust
160 prop_binSR_electron2016M_bin13
161 prop_binCRF_muon2017_bin4
162 prop_binSR_muon2016M_bin21
163 prop_binCRF_muon2018_bin8
164 prop_binCRF_electron2017_bin0
165 prop_binCRF_muon2018_bin9
166 jes_2016M
167 prop_binCROS_muon2017_bin2
168 prop_binCRF_muon2017_bin6
169 prop_binCROS_muon2018_bin7
170 prop_binSR_muon2016M_bin16
171 prop_binCRTT_electron2017_bin0
172 jer_2017
173 prop_binCRF_electron2018_bin7
174 prop_binCROS_electron2017_bin0
175 jer_2018
176 prop_binSR_electron2017_bin18
177 prop_binCROS_electron2017_bin7
178 prop_binSR_muon2016M_bin1
179 prop_binSR_muon2016M_bin20
180 prop_binCRF_muon2016M_bin4
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



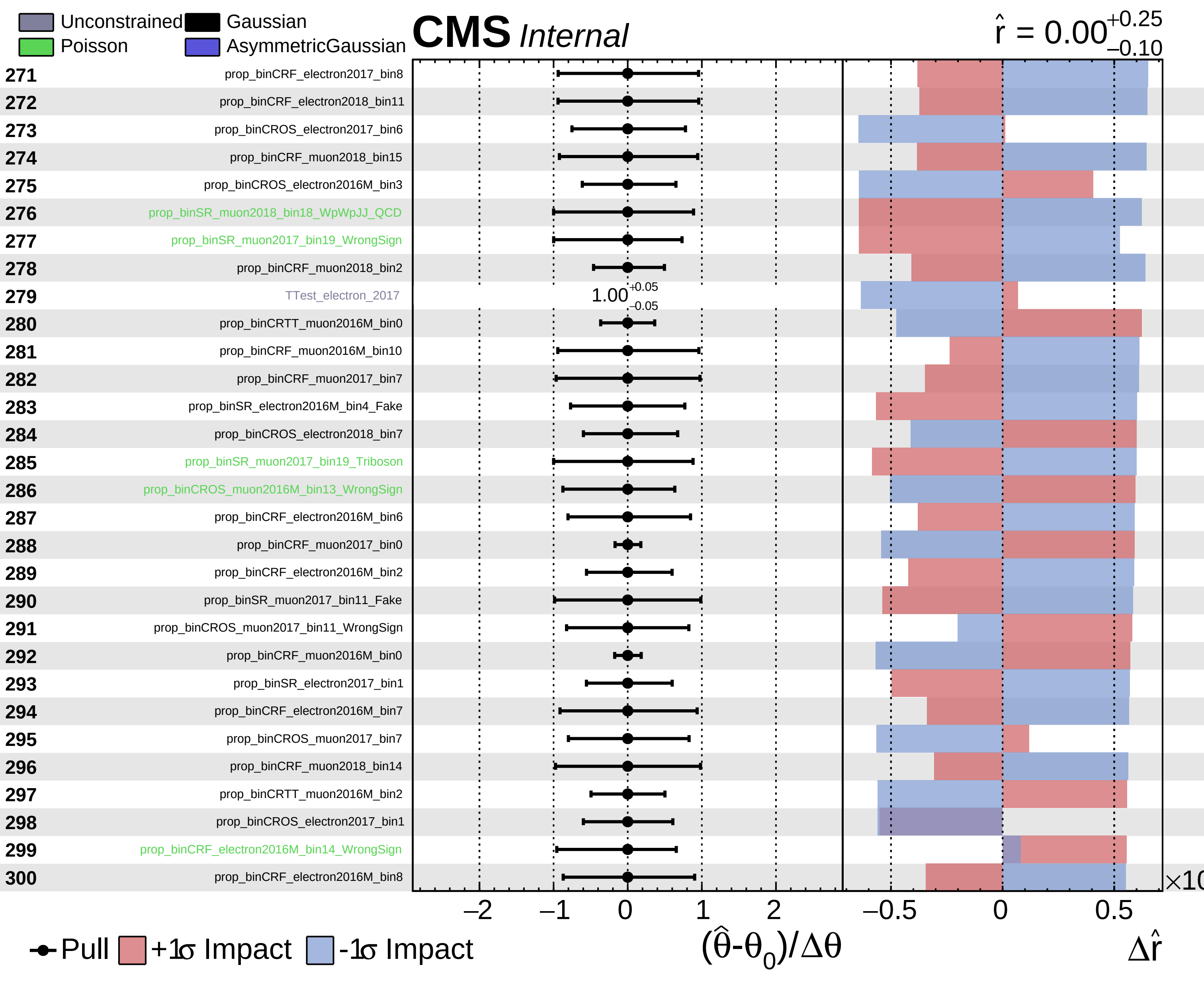
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
181 prop_binCROS_electron2016M_bin8_WrongSign
182 prop_binCROS_electron2016M_bin18_WrongSign
183 prop_binCRF_muon2018_bin12
184 prop_binCROS_muon2016M_bin0
185 prop_binCRF_electron2018_bin3
186 prop_binCROS_electron2016M_bin10
187 tau_vsjet_2017
188 prop_binSR_muon2016M_bin19_TTTo2L2Nu
189 prop_binCRF_muon2016M_bin7
190 prop_binCRF_muon2016M_bin5
191 prop_binSR_electron2016M_bin2
192 prop_binCRF_electron2018_bin6
193 prop_binCROS_electron2016M_bin1
194 prop_binSR_muon2016M_bin12_Fake
195 prop_binSR_electron2016M_bin21_TVX
196 prop_binCRF_electron2018_bin9
197 prop_binCRTT_muon2018_bin4
198 prop_binCRF_muon2018_bin13
199 prop_binCRF_electron2018_bin8
200 prop_binSR_electron2016M_bin1
201 prop_binSR_electron2017_bin20_WZ
202 prop_binCROS_muon2018_bin12_WrongSign
203 prop_binCRF_electron2017_bin4
204 prop_binCROS_muon2017_bin6
205 prop_binCRF_muon2018_bin11
206 prop_binCRF_electron2017_bin2
207 prop_binCROS_muon2017_bin5_WrongSign
208 prop_binCROS_electron2018_bin5
209 prop_binCRF_electron2017_bin3
210 prop_binSR_electron2016M_bin20
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



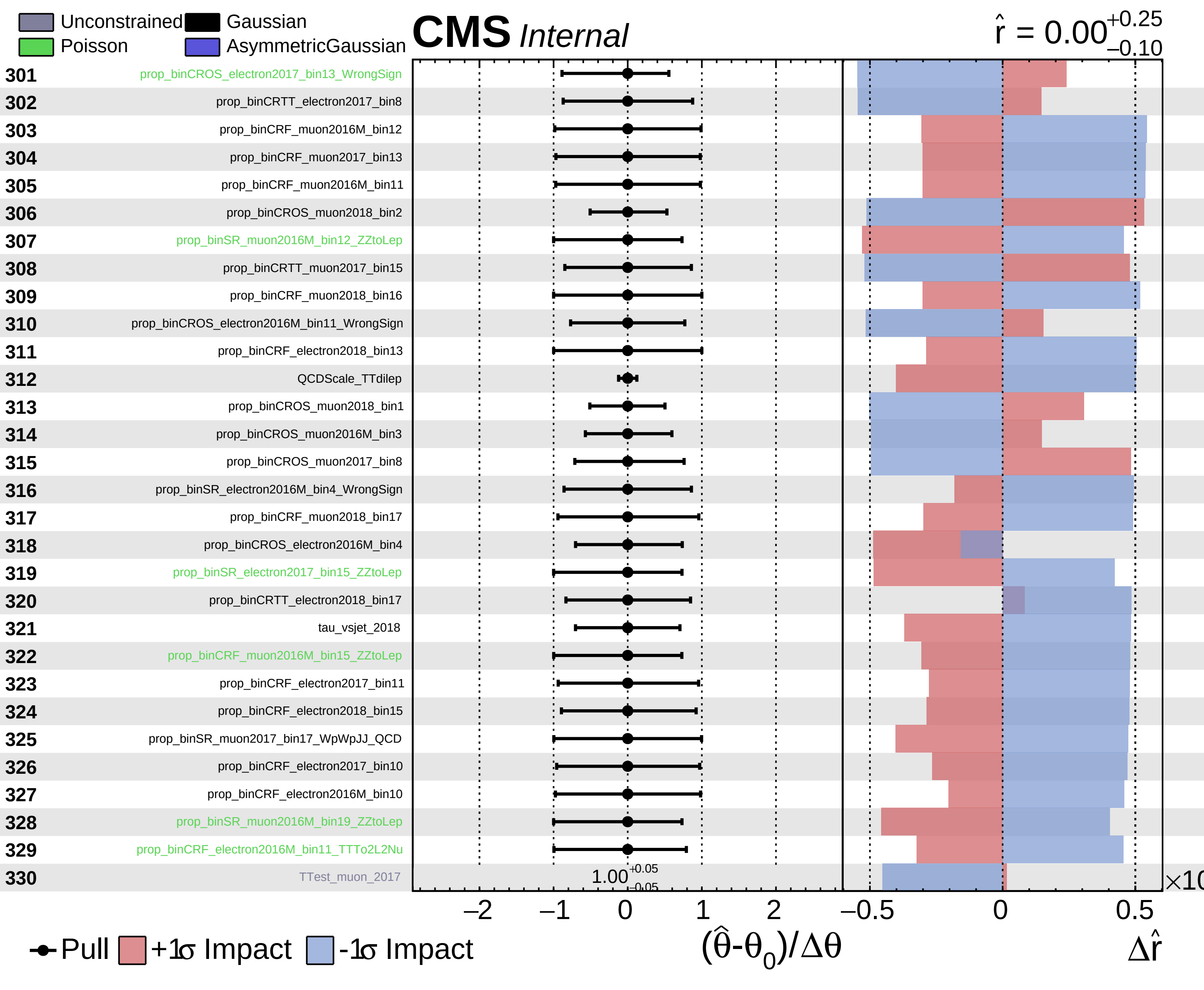
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
211 prop_binCRF_muon2017_bin14_Fake
212 prop_binCROS_electron2017_bin12_WrongSign
213 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
214 prop_binCRF_muon2017_bin8
215 prop_binCRF_muon2016M_bin6
216 jes_2018
217 prop_binCROS_muon2016M_bin5
218 prop_binCRF_electron2017_bin5
219 prop_binCRF_electron2018_bin2
220 prop_binCROS_electron2017_bin8_WrongSign
221 prop_binCRF_electron2017_bin7
222 prop_binCROS_muon2018_bin10
223 prop_binSR_muon2017_bin20_TVX
224 pu
225 prop_binSR_muon2017_bin21
226 prop_binSR_muon2018_bin18_Triboson
227 prop_binSR_muon2018_bin18_TVX
228 prop_binCROS_muon2017_bin14_WrongSign
229 prop_binCRTT_electron2017_bin3
230 prop_binCRF_electron2016M_bin0
231 prop_binCRF_muon2016M_bin8
232 prop_binCRF_muon2016M_bin3
233 prop_binCRF_muon2018_bin10
234 prop_binCROS_muon2016M_bin11
235 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
236 tau_vsjet_2016M
237 prop_binCRTT_muon2016M_bin3
238 prop_binCRF_electron2017_bin15_TVX
239 prop_binCRF_muon2017_bin10
240 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



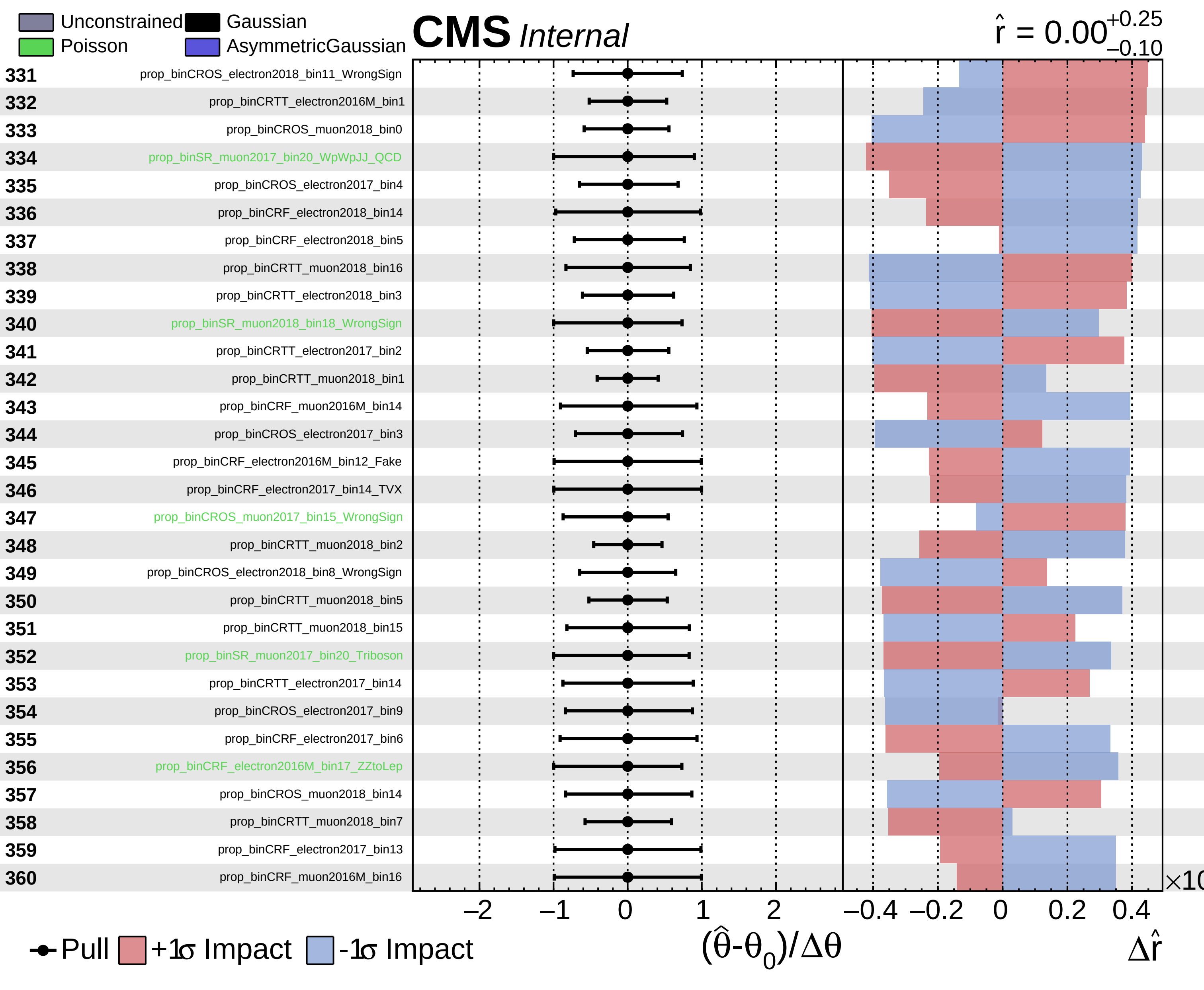
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.10
241 prop_binCROS_electron2016M_bin7_WrongSign
242 prop_binCRTT_muon2017_bin2
243 prop_binCRF_muon2017_bin11
244 prop_binSR_muon2017_bin19_WpWpJJ_QCD
245 prop_binCRF_electron2018_bin1
246 prop_binSR_electron2018_bin2
247 prop_binCRTT_muon2017_bin3
248 prop_binCRF_muon2016M_bin2
249 prop_binCROS_electron2016M_bin5
250 prop_binSR_electron2017_bin2
251 prop_binCROS_electron2018_bin6
252 prop_binCRTT_electron2018_bin5
253 prop_binCRF_electron2016M_bin4
254 prop_binCRF_electron2018_bin12
255 prop_binCRF_electron2018_bin10
256 prop_binCRTT_electron2016M_bin0
257 prop_binCRF_electron2016M_bin5
258 prop_binCRF_electron2017_bin14_Fake
259 prop_binCROS_electron2017_bin5
260 prop_binCRTT_muon2018_bin3
261 prop_binCRF_muon2017_bin9
262 prop_binSR_electron2016M_bin3
263 TTest_muon_2018
1.00 +0.04 −0.04
264 QCDScale_VG
265 prop_binCRTT_muon2018_bin0
266 prop_binCRF_electron2016M_bin3
267 prop_binCROS_muon2018_bin11
268 prop_binCRF_muon2017_bin12
269 prop_binCRF_electron2017_bin9
270 prop_binCRF_muon2016M_bin9
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

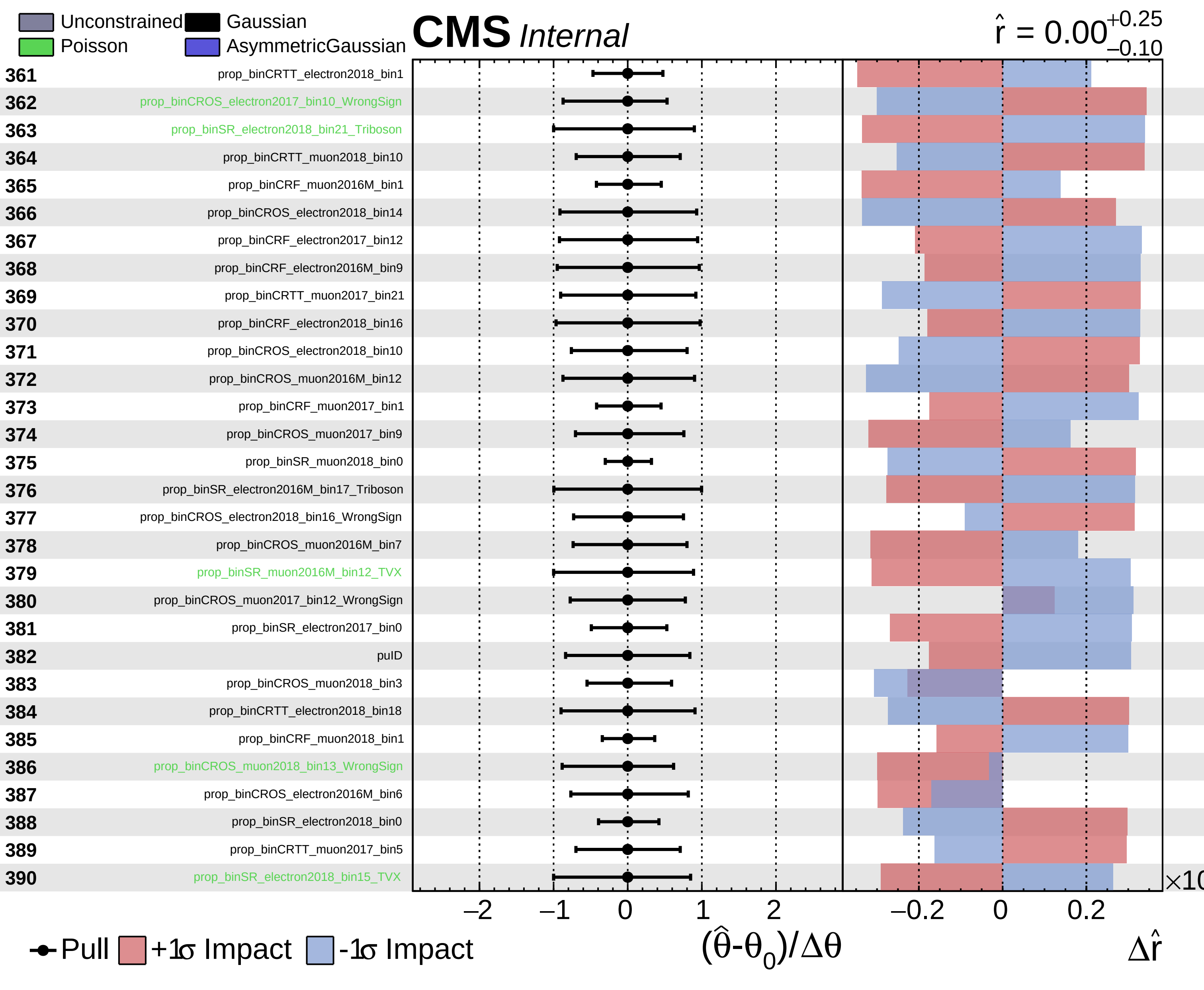


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
271 prop_binCRF_electron2017_bin8
272 prop_binCRF_electron2018_bin11
273 prop_binCROS_electron2017_bin6
274 prop_binCRF_muon2018_bin15
275 prop_binCROS_electron2016M_bin3
276 prop_binSR_muon2018_bin18_WpWpJJ_QCD
277 prop_binSR_muon2017_bin19_WrongSign
278 prop_binCRF_muon2018_bin2
279 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
280 prop_binCRTT_muon2016M_bin0
281 prop_binCRF_muon2016M_bin10
282 prop_binCRF_muon2017_bin7
283 prop_binSR_electron2016M_bin4_Fake
284 prop_binCROS_electron2018_bin7
285 prop_binSR_muon2017_bin19_Triboson
286 prop_binCROS_muon2016M_bin13_WrongSign
287 prop_binCRF_electron2016M_bin6
288 prop_binCRF_muon2017_bin0
289 prop_binCRF_electron2016M_bin2
290 prop_binSR_muon2017_bin11_Fake
291 prop_binCROS_muon2017_bin11_WrongSign
292 prop_binCRF_muon2016M_bin0
293 prop_binSR_electron2017_bin1
294 prop_binCRF_electron2016M_bin7
295 prop_binCROS_muon2017_bin7
296 prop_binCRF_muon2018_bin14
297 prop_binCRTT_muon2016M_bin2
298 prop_binCROS_electron2017_bin1
299 prop_binCRF_electron2016M_bin14_WrongSign
300 prop_binCRF_electron2016M_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

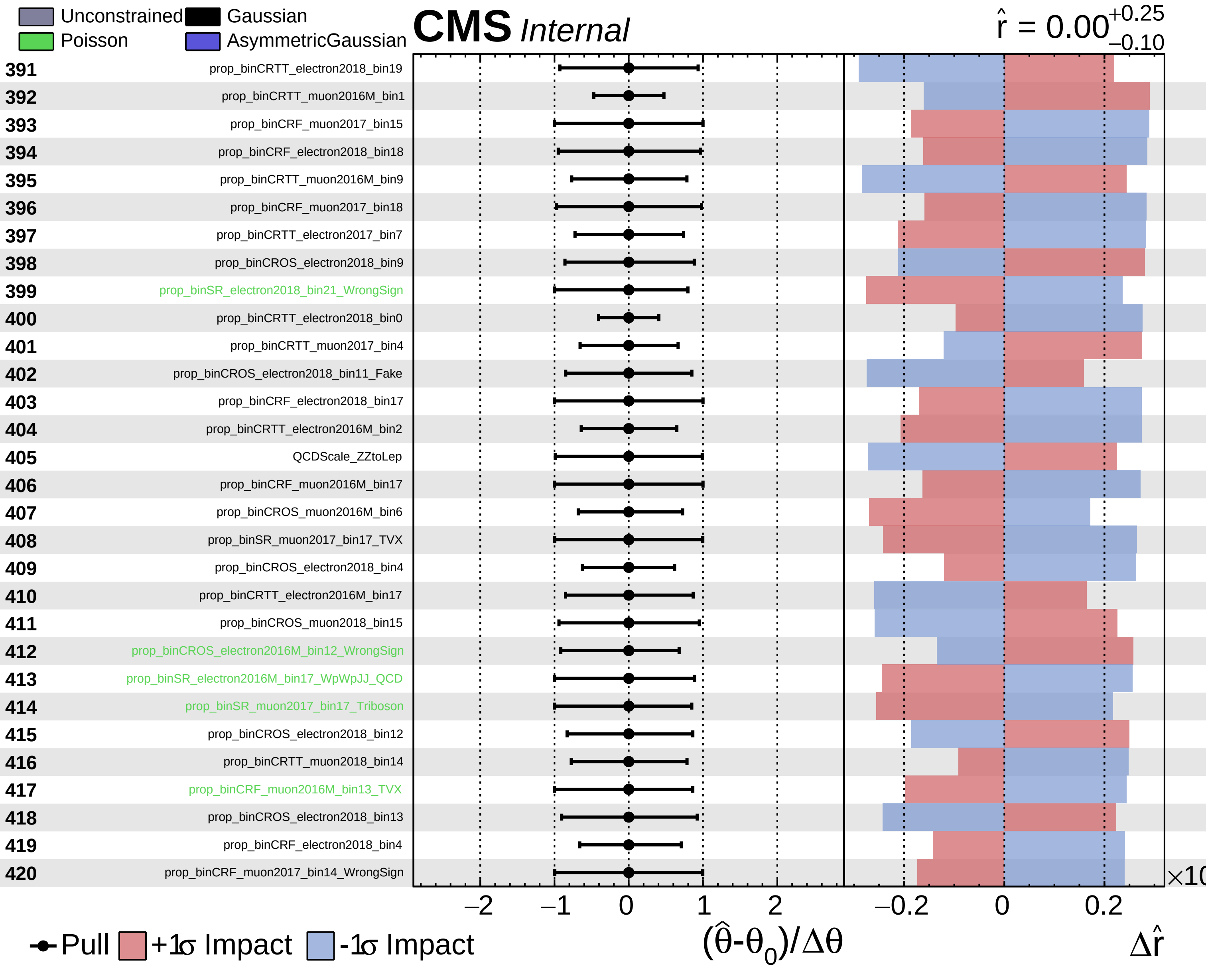


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
301 prop_binCROS_electron2017_bin13_WrongSign
302 prop_binCRTT_electron2017_bin8
303 prop_binCRF_muon2016M_bin12
304 prop_binCRF_muon2017_bin13
305 prop_binCRF_muon2016M_bin11
306 prop_binCROS_muon2018_bin2
307 prop_binSR_muon2016M_bin12_ZZtoLep
308 prop_binCRTT_muon2017_bin15
309 prop_binCRF_muon2018_bin16
310 prop_binCROS_electron2016M_bin11_WrongSign
311 prop_binCRF_electron2018_bin13
312 QCDScale_TTdilep
313 prop_binCROS_muon2018_bin1
314 prop_binCROS_muon2016M_bin3
315 prop_binCROS_muon2017_bin8
316 prop_binSR_electron2016M_bin4_WrongSign
317 prop_binCRF_muon2018_bin17
318 prop_binCROS_electron2016M_bin4
319 prop_binSR_electron2017_bin15_ZZtoLep
320 prop_binCRTT_electron2018_bin17
321 tau_vsjet_2018
322 prop_binCRF_muon2016M_bin15_ZZtoLep
323 prop_binCRF_electron2017_bin11
324 prop_binCRF_electron2018_bin15
325 prop_binSR_muon2017_bin17_WpWpJJ_QCD
326 prop_binCRF_electron2017_bin10
327 prop_binCRF_electron2016M_bin10
328 prop_binSR_muon2016M_bin19_ZZtoLep
329 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
330 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

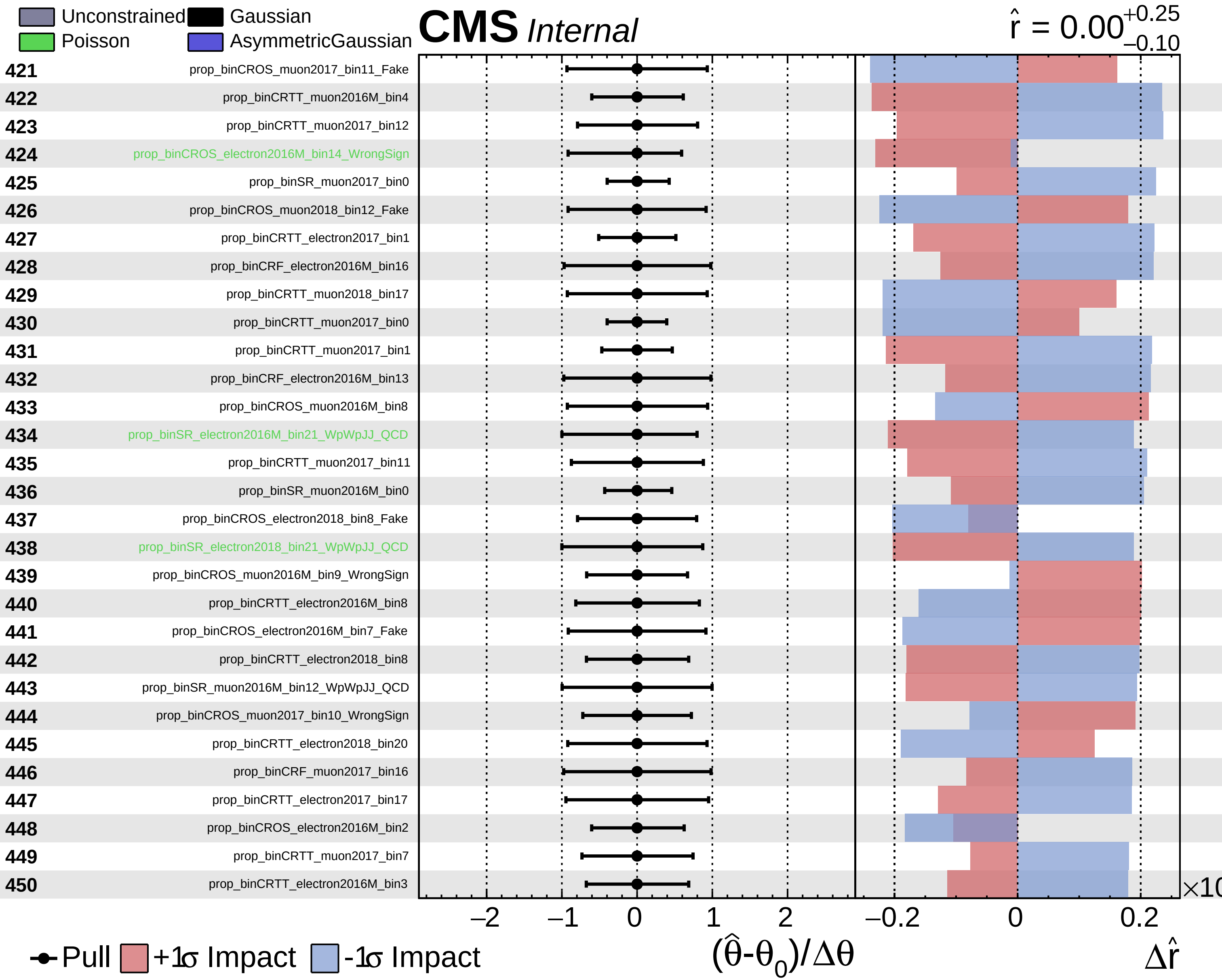

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
331 prop_binCROS_electron2018_bin11_WrongSign
332 prop_binCRTT_electron2016M_bin1
333 prop_binCROS_muon2018_bin0
334 prop_binSR_muon2017_bin20_WpWpJJ_QCD
335 prop_binCROS_electron2017_bin4
336 prop_binCRF_electron2018_bin14
337 prop_binCRF_electron2018_bin5
338 prop_binCRTT_muon2018_bin16
339 prop_binCRTT_electron2018_bin3
340 prop_binSR_muon2018_bin18_WrongSign
341 prop_binCRTT_electron2017_bin2
342 prop_binCRTT_muon2018_bin1
343 prop_binCRF_muon2016M_bin14
344 prop_binCROS_electron2017_bin3
345 prop_binCRF_electron2016M_bin12_Fake
346 prop_binCRF_electron2017_bin14_TVX
347 prop_binCROS_muon2017_bin15_WrongSign
348 prop_binCRTT_muon2018_bin2
349 prop_binCROS_electron2018_bin8_WrongSign
350 prop_binCRTT_muon2018_bin5
351 prop_binCRTT_muon2018_bin15
352 prop_binSR_muon2017_bin20_Triboson
353 prop_binCRTT_electron2017_bin14
354 prop_binCROS_electron2017_bin9
355 prop_binCRF_electron2017_bin6
356 prop_binCRF_electron2016M_bin17_ZZtoLep
357 prop_binCROS_muon2018_bin14
358 prop_binCRTT_muon2018_bin7
359 prop_binCRF_electron2017_bin13
360 prop_binCRF_muon2016M_bin16
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



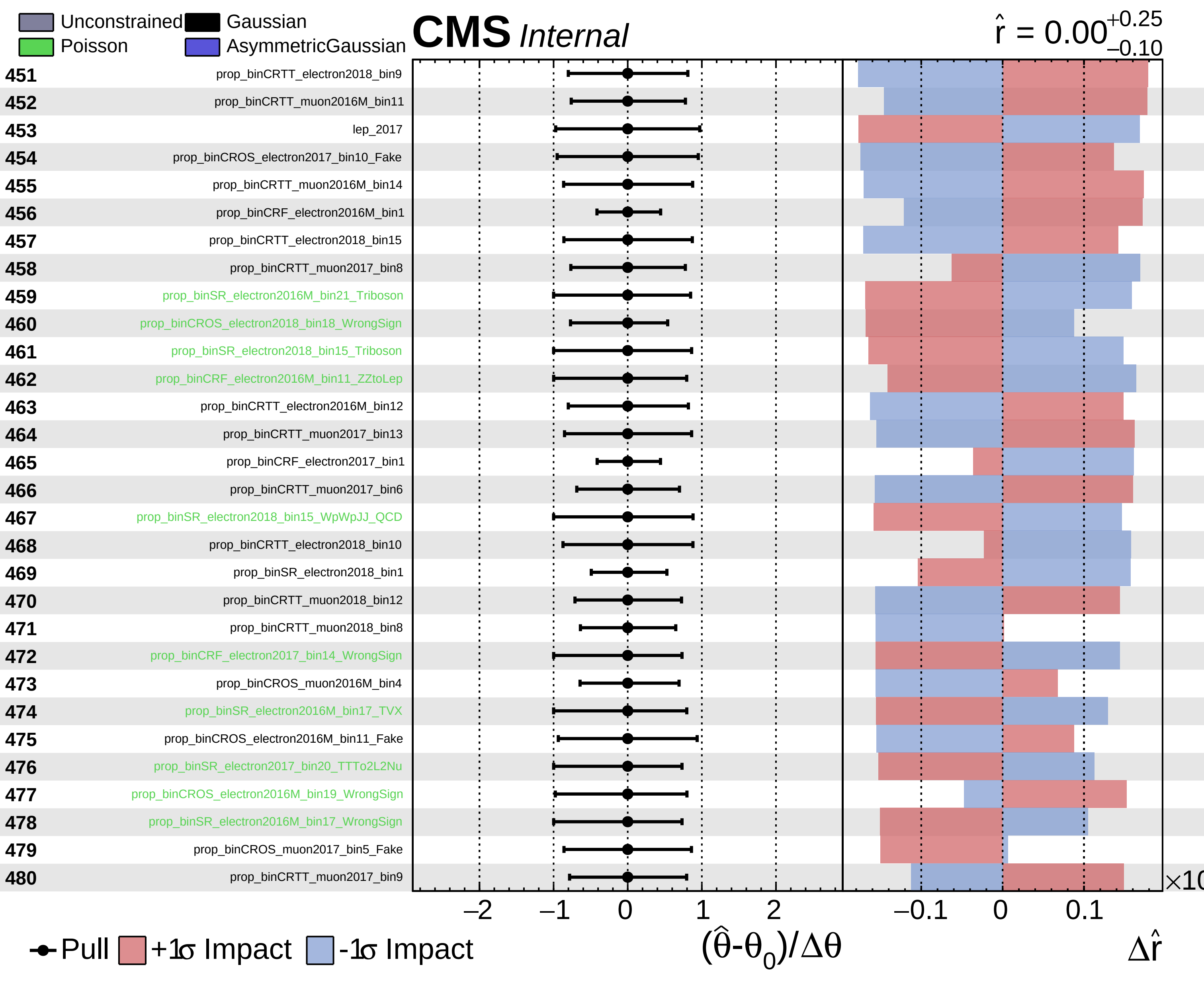
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
361 prop_binCRTT_electron2018_bin1
362 prop_binCROS_electron2017_bin10_WrongSign
363 prop_binSR_electron2018_bin21_Triboson
364 prop_binCRTT_muon2018_bin10
365 prop_binCRF_muon2016M_bin1
366 prop_binCROS_electron2018_bin14
367 prop_binCRF_electron2017_bin12
368 prop_binCRF_electron2016M_bin9
369 prop_binCRTT_muon2017_bin21
370 prop_binCRF_electron2018_bin16
371 prop_binCROS_electron2018_bin10
372 prop_binCROS_muon2016M_bin12
373 prop_binCRF_muon2017_bin1
374 prop_binCROS_muon2017_bin9
375 prop_binSR_muon2018_bin0
376 prop_binSR_electron2016M_bin17_Triboson
377 prop_binCROS_electron2018_bin16_WrongSign
378 prop_binCROS_muon2016M_bin7
379 prop_binSR_muon2016M_bin12_TVX
380 prop_binCROS_muon2017_bin12_WrongSign
381 prop_binSR_electron2017_bin0
382 puID
383 prop_binCROS_muon2018_bin3
384 prop_binCRTT_electron2018_bin18
385 prop_binCRF_muon2018_bin1
386 prop_binCROS_muon2018_bin13_WrongSign
387 prop_binCROS_electron2016M_bin6
388 prop_binSR_electron2018_bin0
389 prop_binCRTT_muon2017_bin5
390 prop_binSR_electron2018_bin15_TVX
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



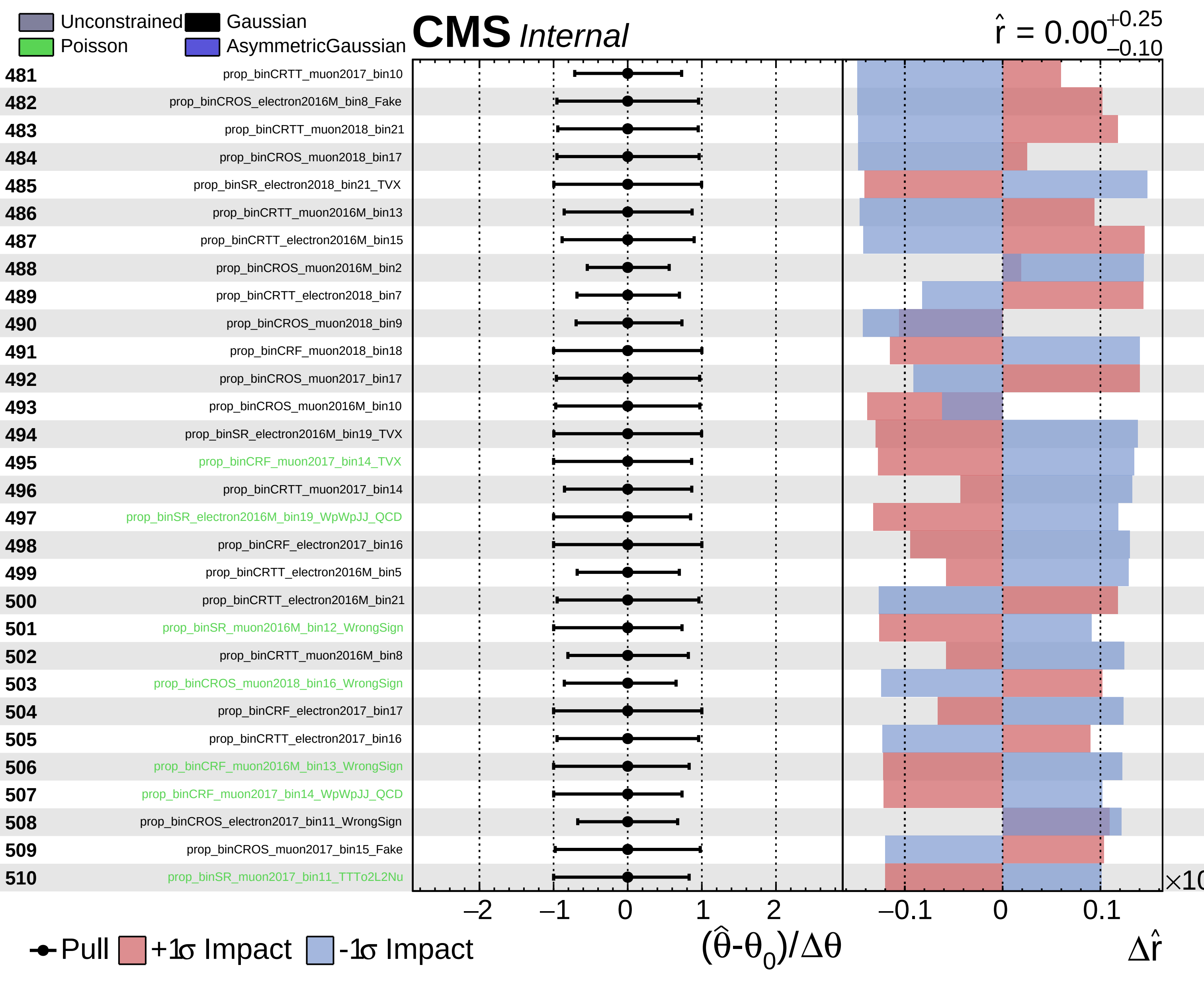
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
391 prop_binCRTT_electron2018_bin19
392 prop_binCRTT_muon2016M_bin1
393 prop_binCRF_muon2017_bin15
394 prop_binCRF_electron2018_bin18
395 prop_binCRTT_muon2016M_bin9
396 prop_binCRF_muon2017_bin18
397 prop_binCRTT_electron2017_bin7
398 prop_binCROS_electron2018_bin9
399 prop_binSR_electron2018_bin21_WrongSign
400 prop_binCRTT_electron2018_bin0
401 prop_binCRTT_muon2017_bin4
402 prop_binCROS_electron2018_bin11_Fake
403 prop_binCRF_electron2018_bin17
404 prop_binCRTT_electron2016M_bin2
405 QCDScale_ZZtoLep
406 prop_binCRF_muon2016M_bin17
407 prop_binCROS_muon2016M_bin6
408 prop_binSR_muon2017_bin17_TVX
409 prop_binCROS_electron2018_bin4
410 prop_binCRTT_electron2016M_bin17
411 prop_binCROS_muon2018_bin15
412 prop_binCROS_electron2016M_bin12_WrongSign
413 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
414 prop_binSR_muon2017_bin17_Triboson
415 prop_binCROS_electron2018_bin12
416 prop_binCRTT_muon2018_bin14
417 prop_binCRF_muon2016M_bin13_TVX
418 prop_binCROS_electron2018_bin13
419 prop_binCRF_electron2018_bin4
420 prop_binCRF_muon2017_bin14_WrongSign
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



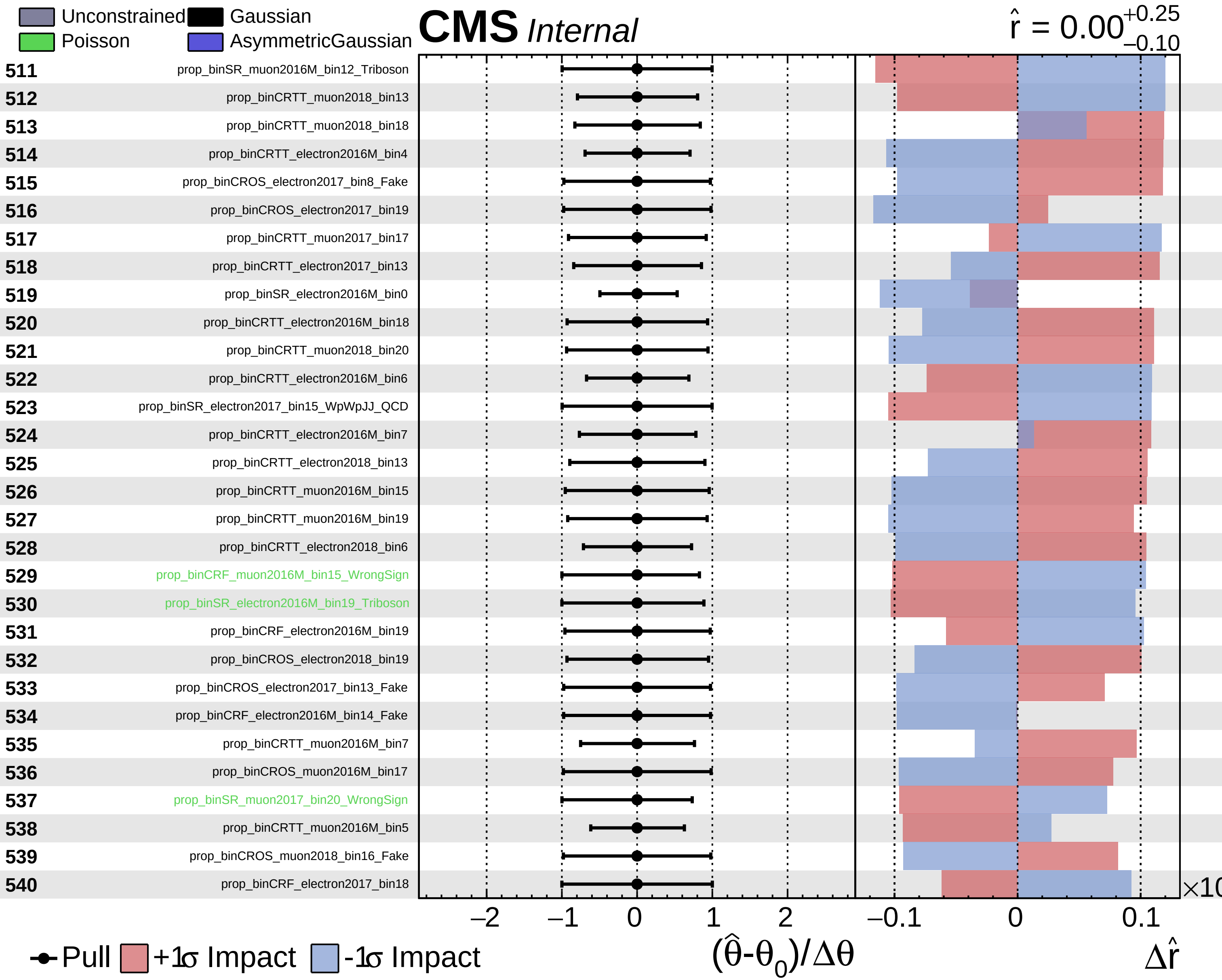
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
421 prop_binCROS_muon2017_bin11_Fake
422 prop_binCRTT_muon2016M_bin4
423 prop_binCRTT_muon2017_bin12
424 prop_binCROS_electron2016M_bin14_WrongSign
425 prop_binSR_muon2017_bin0
426 prop_binCROS_muon2018_bin12_Fake
427 prop_binCRTT_electron2017_bin1
428 prop_binCRF_electron2016M_bin16
429 prop_binCRTT_muon2018_bin17
430 prop_binCRTT_muon2017_bin0
431 prop_binCRTT_muon2017_bin1
432 prop_binCRF_electron2016M_bin13
433 prop_binCROS_muon2016M_bin8
434 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
435 prop_binCRTT_muon2017_bin11
436 prop_binSR_muon2016M_bin0
437 prop_binCROS_electron2018_bin8_Fake
438 prop_binSR_electron2018_bin21_WpWpJJ_QCD
439 prop_binCROS_muon2016M_bin9_WrongSign
440 prop_binCRTT_electron2016M_bin8
441 prop_binCROS_electron2016M_bin7_Fake
442 prop_binCRTT_electron2018_bin8
443 prop_binSR_muon2016M_bin12_WpWpJJ_QCD
444 prop_binCROS_muon2017_bin10_WrongSign
445 prop_binCRTT_electron2018_bin20
446 prop_binCRF_muon2017_bin16
447 prop_binCRTT_electron2017_bin17
448 prop_binCROS_electron2016M_bin2
449 prop_binCRTT_muon2017_bin7
450 prop_binCRTT_electron2016M_bin3
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



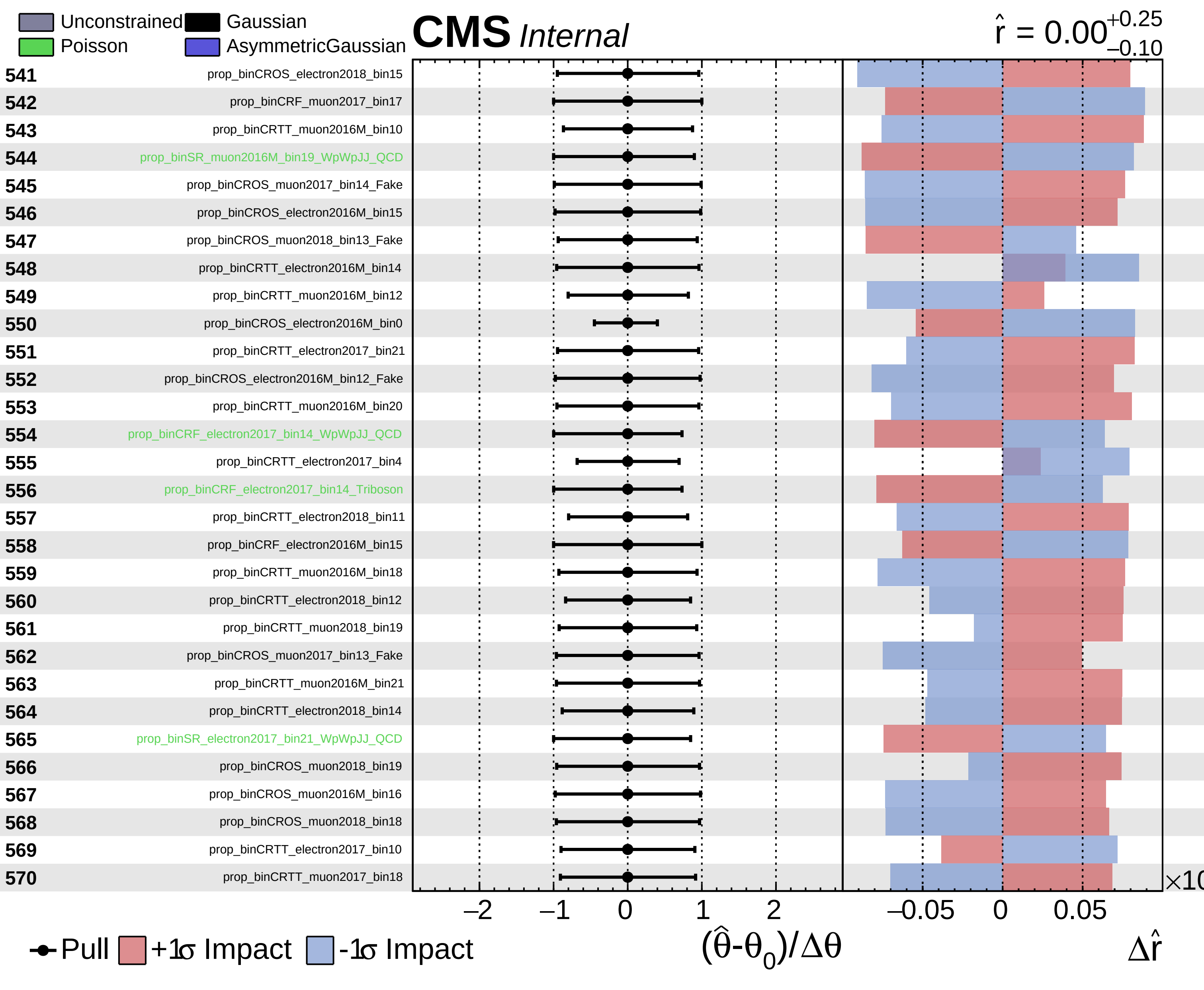
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
451 prop_binCRTT_electron2018_bin9
452 prop_binCRTT_muon2016M_bin11
453 lep_2017
454 prop_binCROS_electron2017_bin10_Fake
455 prop_binCRTT_muon2016M_bin14
456 prop_binCRF_electron2016M_bin1
457 prop_binCRTT_electron2018_bin15
458 prop_binCRTT_muon2017_bin8
459 prop_binSR_electron2016M_bin21_Triboson
460 prop_binCROS_electron2018_bin18_WrongSign
461 prop_binSR_electron2018_bin15_Triboson
462 prop_binCRF_electron2016M_bin11_ZZtoLep
463 prop_binCRTT_electron2016M_bin12
464 prop_binCRTT_muon2017_bin13
465 prop_binCRF_electron2017_bin1
466 prop_binCRTT_muon2017_bin6
467 prop_binSR_electron2018_bin15_WpWpJJ_QCD
468 prop_binCRTT_electron2018_bin10
469 prop_binSR_electron2018_bin1
470 prop_binCRTT_muon2018_bin12
471 prop_binCRTT_muon2018_bin8
472 prop_binCRF_electron2017_bin14_WrongSign
473 prop_binCROS_muon2016M_bin4
474 prop_binSR_electron2016M_bin17_TVX
475 prop_binCROS_electron2016M_bin11_Fake
476 prop_binSR_electron2017_bin20_TTTo2L2Nu
477 prop_binCROS_electron2016M_bin19_WrongSign
478 prop_binSR_electron2016M_bin17_WrongSign
479 prop_binCROS_muon2017_bin5_Fake
480 prop_binCRTT_muon2017_bin9
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



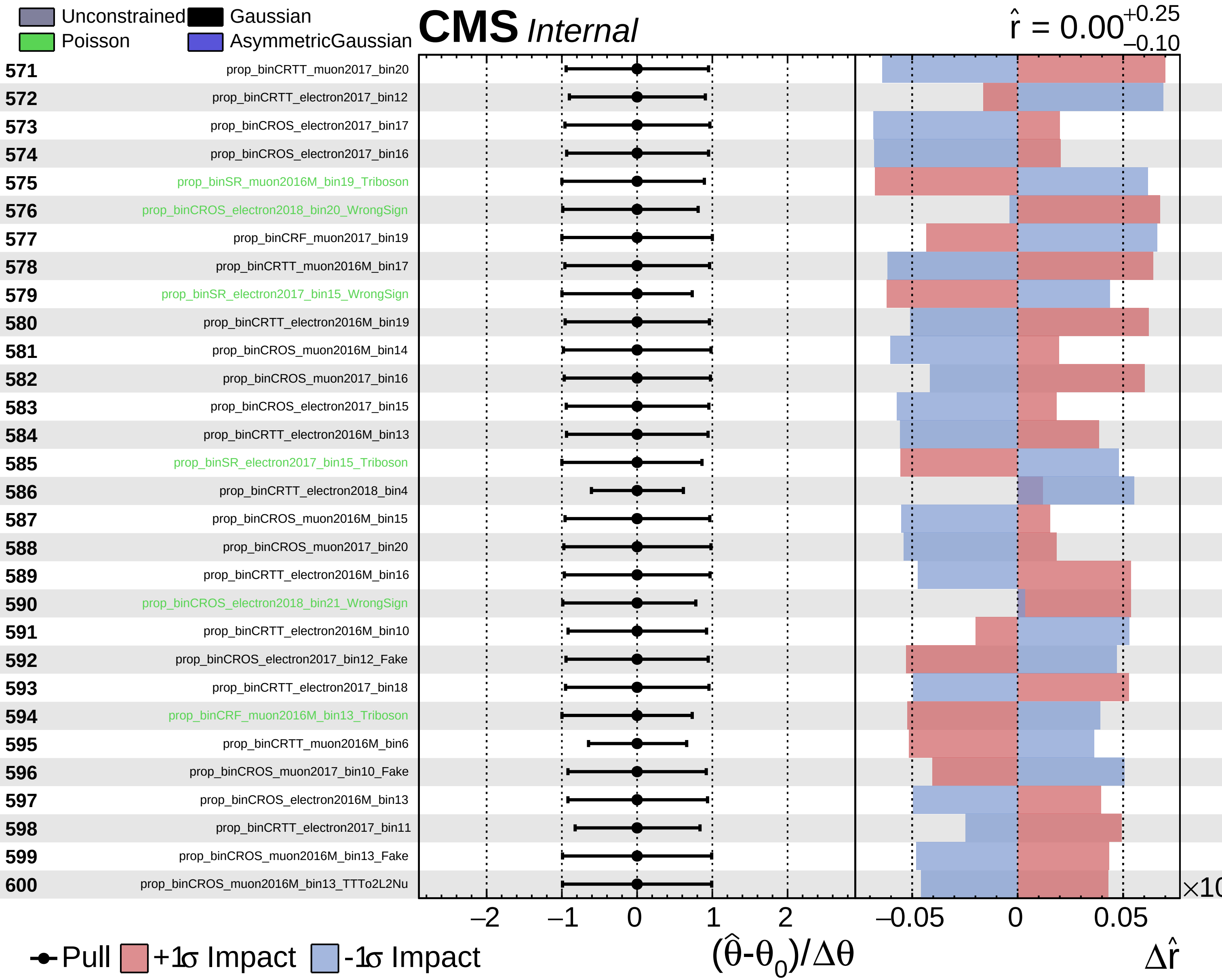
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
481 prop_binCRTT_muon2017_bin10
482 prop_binCROS_electron2016M_bin8_Fake
483 prop_binCRTT_muon2018_bin21
484 prop_binCROS_muon2018_bin17
485 prop_binSR_electron2018_bin21_TVX
486 prop_binCRTT_muon2016M_bin13
487 prop_binCRTT_electron2016M_bin15
488 prop_binCROS_muon2016M_bin2
489 prop_binCRTT_electron2018_bin7
490 prop_binCROS_muon2018_bin9
491 prop_binCRF_muon2018_bin18
492 prop_binCROS_muon2017_bin17
493 prop_binCROS_muon2016M_bin10
494 prop_binSR_electron2016M_bin19_TVX
495 prop_binCRF_muon2017_bin14_TVX
496 prop_binCRTT_muon2017_bin14
497 prop_binSR_electron2016M_bin19_WpWpJJ_QCD
498 prop_binCRF_electron2017_bin16
499 prop_binCRTT_electron2016M_bin5
500 prop_binCRTT_electron2016M_bin21
501 prop_binSR_muon2016M_bin12_WrongSign
502 prop_binCRTT_muon2016M_bin8
503 prop_binCROS_muon2018_bin16_WrongSign
504 prop_binCRF_electron2017_bin17
505 prop_binCRTT_electron2017_bin16
506 prop_binCRF_muon2016M_bin13_WrongSign
507 prop_binCRF_muon2017_bin14_WpWpJJ_QCD
508 prop_binCROS_electron2017_bin11_WrongSign
509 prop_binCROS_muon2017_bin15_Fake
510 prop_binSR_muon2017_bin11_TTTo2L2Nu
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



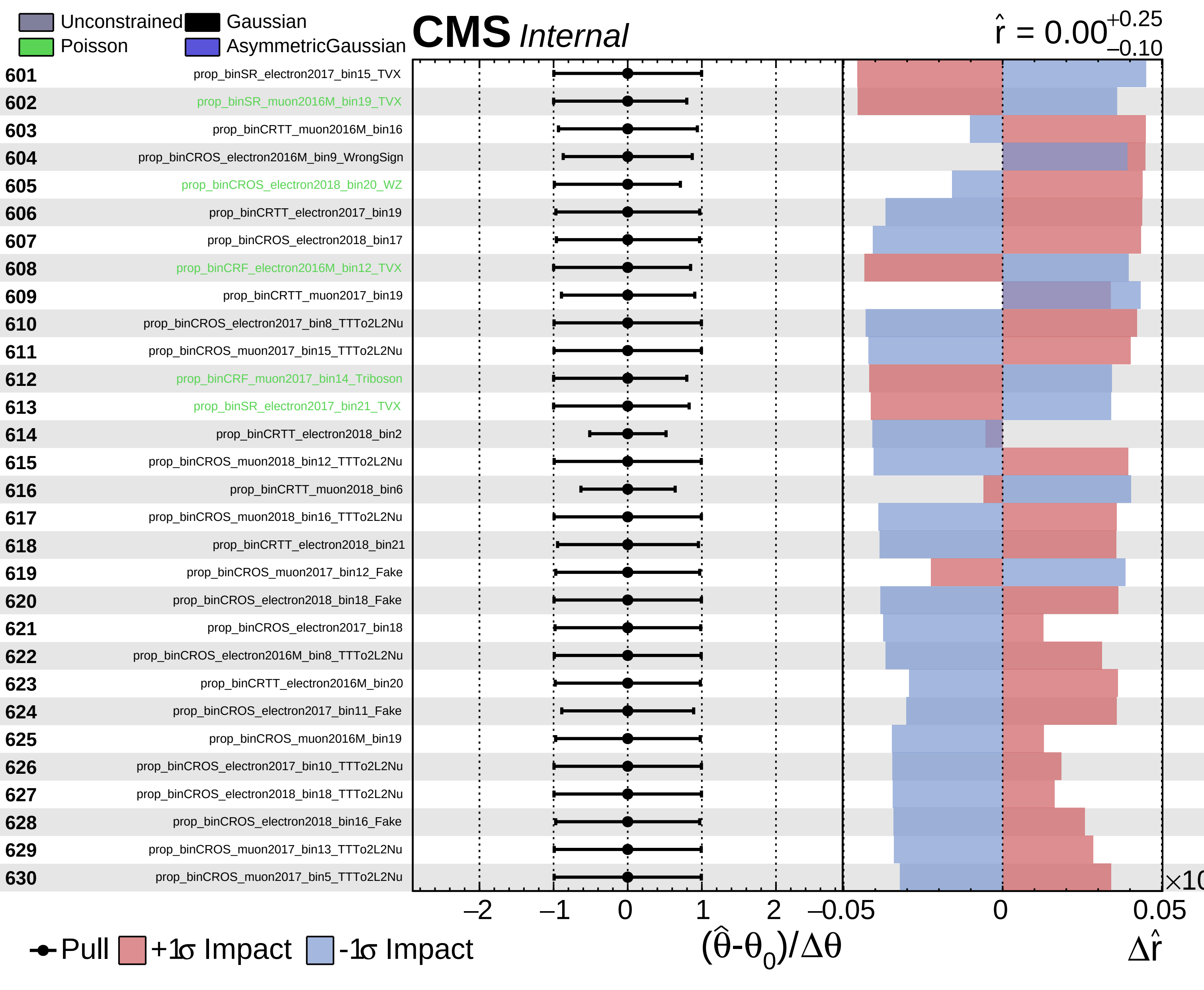
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
511 prop_binSR_muon2016M_bin12_Triboson
512 prop_binCRTT_muon2018_bin13
513 prop_binCRTT_muon2018_bin18
514 prop_binCRTT_electron2016M_bin4
515 prop_binCROS_electron2017_bin8_Fake
516 prop_binCROS_electron2017_bin19
517 prop_binCRTT_muon2017_bin17
518 prop_binCRTT_electron2017_bin13
519 prop_binSR_electron2016M_bin0
520 prop_binCRTT_electron2016M_bin18
521 prop_binCRTT_muon2018_bin20
522 prop_binCRTT_electron2016M_bin6
523 prop_binSR_electron2017_bin15_WpWpJJ_QCD
524 prop_binCRTT_electron2016M_bin7
525 prop_binCRTT_electron2018_bin13
526 prop_binCRTT_muon2016M_bin15
527 prop_binCRTT_muon2016M_bin19
528 prop_binCRTT_electron2018_bin6
529 prop_binCRF_muon2016M_bin15_WrongSign
530 prop_binSR_electron2016M_bin19_Triboson
531 prop_binCRF_electron2016M_bin19
532 prop_binCROS_electron2018_bin19
533 prop_binCROS_electron2017_bin13_Fake
534 prop_binCRF_electron2016M_bin14_Fake
535 prop_binCRTT_muon2016M_bin7
536 prop_binCROS_muon2016M_bin17
537 prop_binSR_muon2017_bin20_WrongSign
538 prop_binCRTT_muon2016M_bin5
539 prop_binCROS_muon2018_bin16_Fake
540 prop_binCRF_electron2017_bin18
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



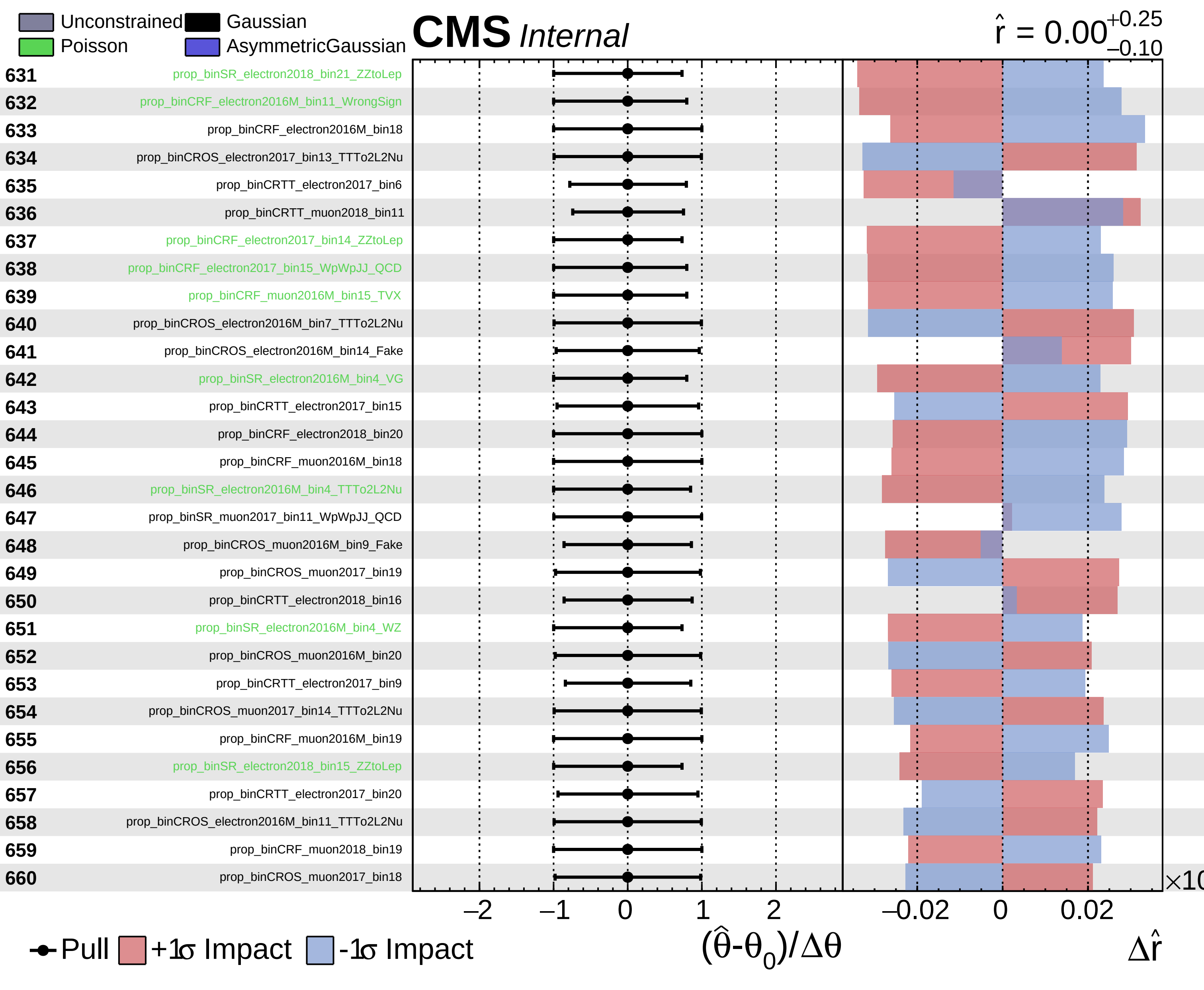
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
541 prop_binCROS_electron2018_bin15
542 prop_binCRF_muon2017_bin17
543 prop_binCRTT_muon2016M_bin10
544 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
545 prop_binCROS_muon2017_bin14_Fake
546 prop_binCROS_electron2016M_bin15
547 prop_binCROS_muon2018_bin13_Fake
548 prop_binCRTT_electron2016M_bin14
549 prop_binCRTT_muon2016M_bin12
550 prop_binCROS_electron2016M_bin0
551 prop_binCRTT_electron2017_bin21
552 prop_binCROS_electron2016M_bin12_Fake
553 prop_binCRTT_muon2016M_bin20
554 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
555 prop_binCRTT_electron2017_bin4
556 prop_binCRF_electron2017_bin14_Triboson
557 prop_binCRTT_electron2018_bin11
558 prop_binCRF_electron2016M_bin15
559 prop_binCRTT_muon2016M_bin18
560 prop_binCRTT_electron2018_bin12
561 prop_binCRTT_muon2018_bin19
562 prop_binCROS_muon2017_bin13_Fake
563 prop_binCRTT_muon2016M_bin21
564 prop_binCRTT_electron2018_bin14
565 prop_binSR_electron2017_bin21_WpWpJJ_QCD
566 prop_binCROS_muon2018_bin19
567 prop_binCROS_muon2016M_bin16
568 prop_binCROS_muon2018_bin18
569 prop_binCRTT_electron2017_bin10
570 prop_binCRTT_muon2017_bin18
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



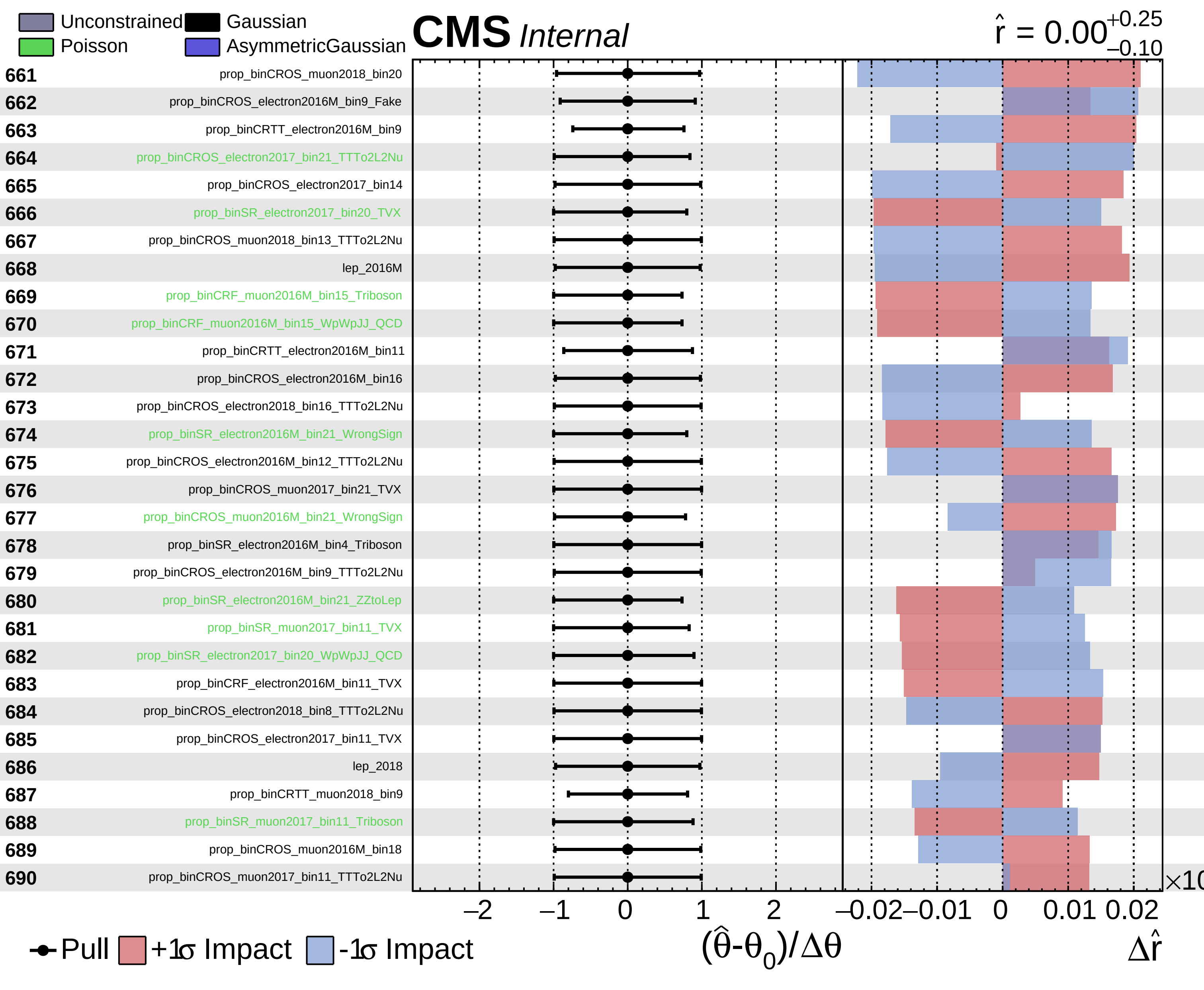
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.10
571 prop_binCRTT_muon2017_bin20
572 prop_binCRTT_electron2017_bin12
573 prop_binCROS_electron2017_bin17
574 prop_binCROS_electron2017_bin16
575 prop_binSR_muon2016M_bin19_Triboson
576 prop_binCROS_electron2018_bin20_WrongSign
577 prop_binCRF_muon2017_bin19
578 prop_binCRTT_muon2016M_bin17
579 prop_binSR_electron2017_bin15_WrongSign
580 prop_binCRTT_electron2016M_bin19
581 prop_binCROS_muon2016M_bin14
582 prop_binCROS_muon2017_bin16
583 prop_binCROS_electron2017_bin15
584 prop_binCRTT_electron2016M_bin13
585 prop_binSR_electron2017_bin15_Triboson
586 prop_binCRTT_electron2018_bin4
587 prop_binCROS_muon2016M_bin15
588 prop_binCROS_muon2017_bin20
589 prop_binCRTT_electron2016M_bin16
590 prop_binCROS_electron2018_bin21_WrongSign
591 prop_binCRTT_electron2016M_bin10
592 prop_binCROS_electron2017_bin12_Fake
593 prop_binCRTT_electron2017_bin18
594 prop_binCRF_muon2016M_bin13_Triboson
595 prop_binCRTT_muon2016M_bin6
596 prop_binCROS_muon2017_bin10_Fake
597 prop_binCROS_electron2016M_bin13
598 prop_binCRTT_electron2017_bin11
599 prop_binCROS_muon2016M_bin13_Fake
600 prop_binCROS_muon2016M_bin13_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



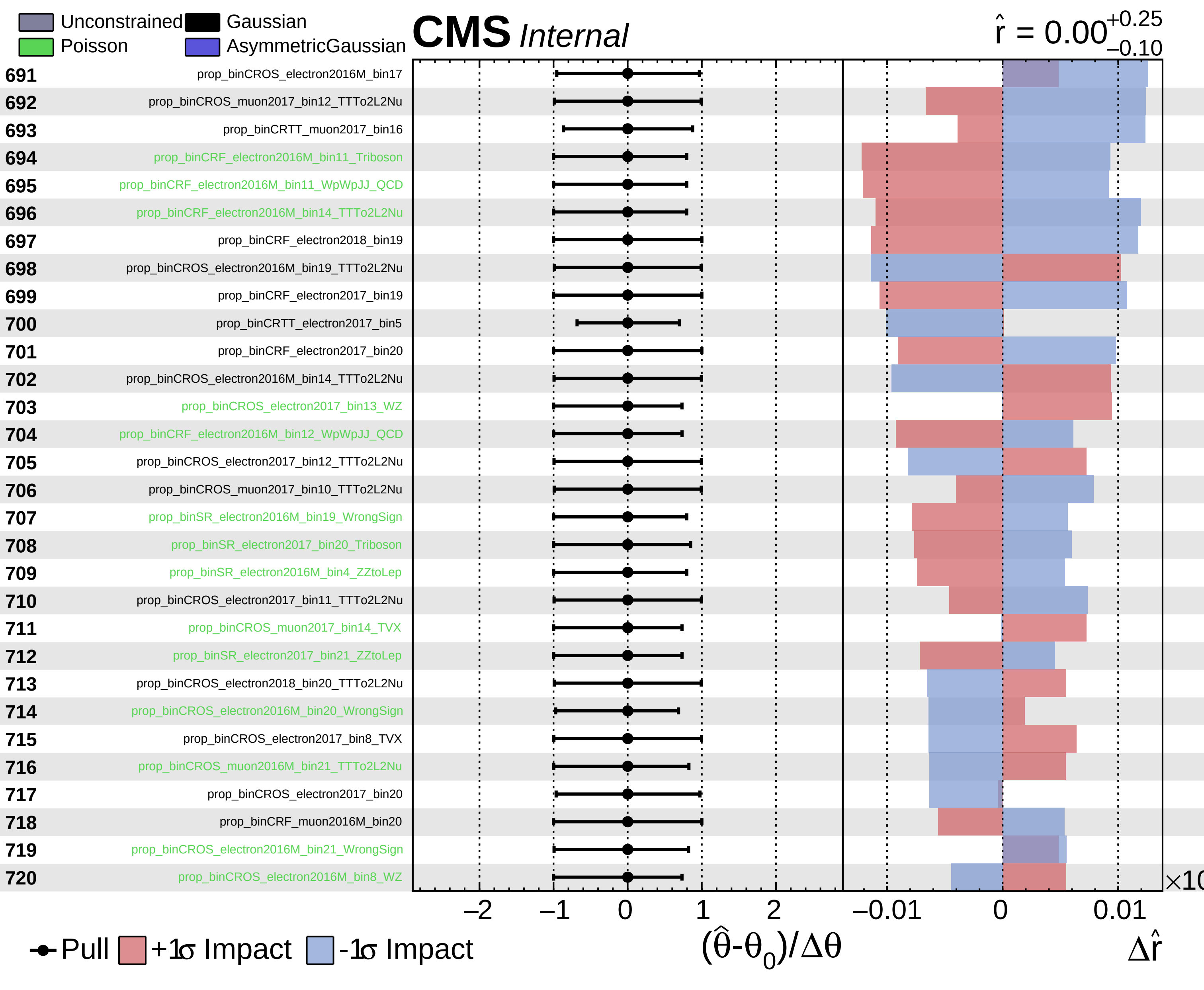
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
601 prop_binSR_electron2017_bin15_TVX
602 prop_binSR_muon2016M_bin19_TVX
603 prop_binCRTT_muon2016M_bin16
604 prop_binCROS_electron2016M_bin9_WrongSign
605 prop_binCROS_electron2018_bin20_WZ
606 prop_binCRTT_electron2017_bin19
607 prop_binCROS_electron2018_bin17
608 prop_binCRF_electron2016M_bin12_TVX
609 prop_binCRTT_muon2017_bin19
610 prop_binCROS_electron2017_bin8_TTTo2L2Nu
611 prop_binCROS_muon2017_bin15_TTTo2L2Nu
612 prop_binCRF_muon2017_bin14_Triboson
613 prop_binSR_electron2017_bin21_TVX
614 prop_binCRTT_electron2018_bin2
615 prop_binCROS_muon2018_bin12_TTTo2L2Nu
616 prop_binCRTT_muon2018_bin6
617 prop_binCROS_muon2018_bin16_TTTo2L2Nu
618 prop_binCRTT_electron2018_bin21
619 prop_binCROS_muon2017_bin12_Fake
620 prop_binCROS_electron2018_bin18_Fake
621 prop_binCROS_electron2017_bin18
622 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
623 prop_binCRTT_electron2016M_bin20
624 prop_binCROS_electron2017_bin11_Fake
625 prop_binCROS_muon2016M_bin19
626 prop_binCROS_electron2017_bin10_TTTo2L2Nu
627 prop_binCROS_electron2018_bin18_TTTo2L2Nu
628 prop_binCROS_electron2018_bin16_Fake
629 prop_binCROS_muon2017_bin13_TTTo2L2Nu
630 prop_binCROS_muon2017_bin5_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



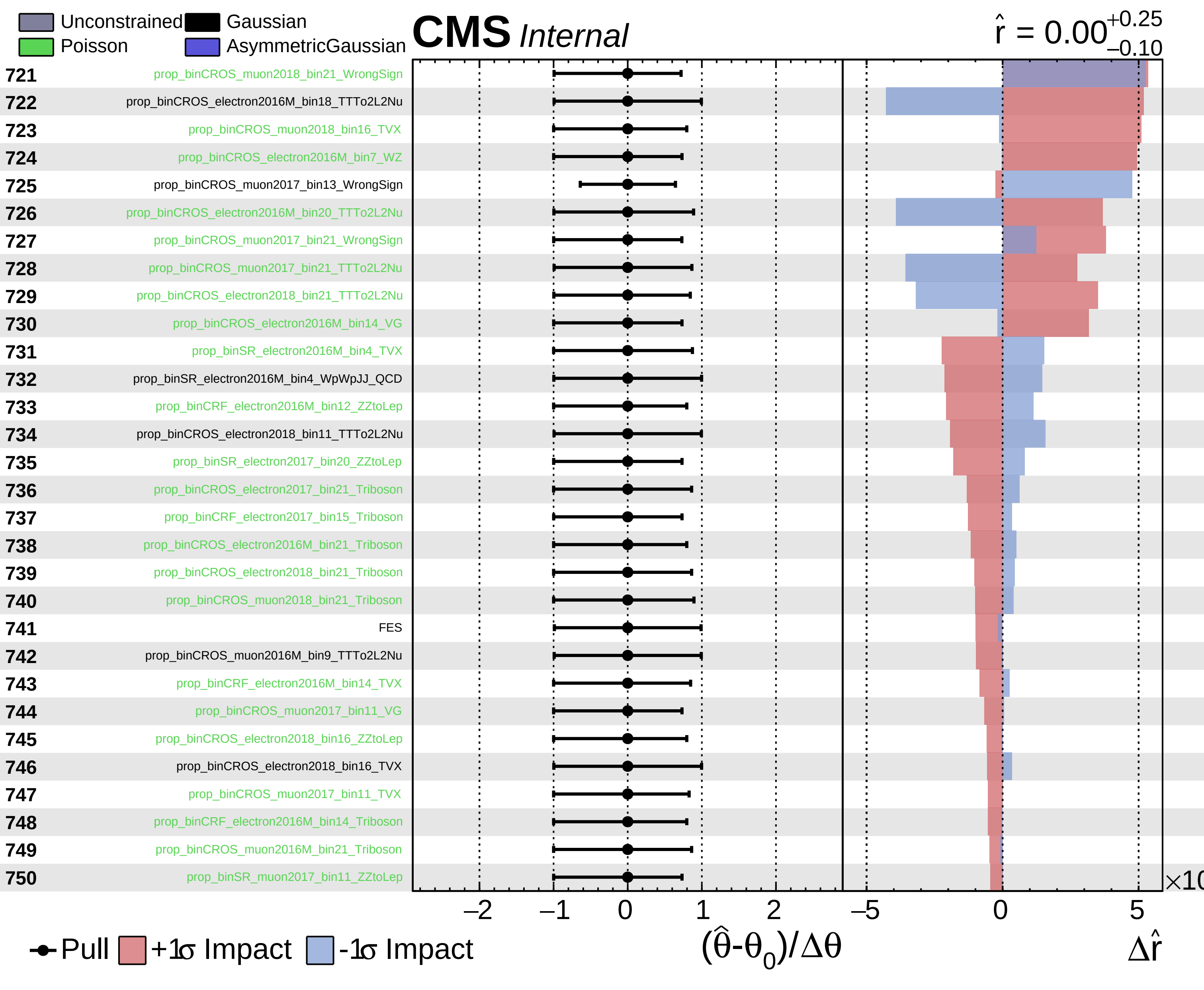
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
631 prop_binSR_electron2018_bin21_ZZtoLep
632 prop_binCRF_electron2016M_bin11_WrongSign
633 prop_binCRF_electron2016M_bin18
634 prop_binCROS_electron2017_bin13_TTTo2L2Nu
635 prop_binCRTT_electron2017_bin6
636 prop_binCRTT_muon2018_bin11
637 prop_binCRF_electron2017_bin14_ZZtoLep
638 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
639 prop_binCRF_muon2016M_bin15_TVX
640 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
641 prop_binCROS_electron2016M_bin14_Fake
642 prop_binSR_electron2016M_bin4_VG
643 prop_binCRTT_electron2017_bin15
644 prop_binCRF_electron2018_bin20
645 prop_binCRF_muon2016M_bin18
646 prop_binSR_electron2016M_bin4_TTTo2L2Nu
647 prop_binSR_muon2017_bin11_WpWpJJ_QCD
648 prop_binCROS_muon2016M_bin9_Fake
649 prop_binCROS_muon2017_bin19
650 prop_binCRTT_electron2018_bin16
651 prop_binSR_electron2016M_bin4_WZ
652 prop_binCROS_muon2016M_bin20
653 prop_binCRTT_electron2017_bin9
654 prop_binCROS_muon2017_bin14_TTTo2L2Nu
655 prop_binCRF_muon2016M_bin19
656 prop_binSR_electron2018_bin15_ZZtoLep
657 prop_binCRTT_electron2017_bin20
658 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
659 prop_binCRF_muon2018_bin19
660 prop_binCROS_muon2017_bin18
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

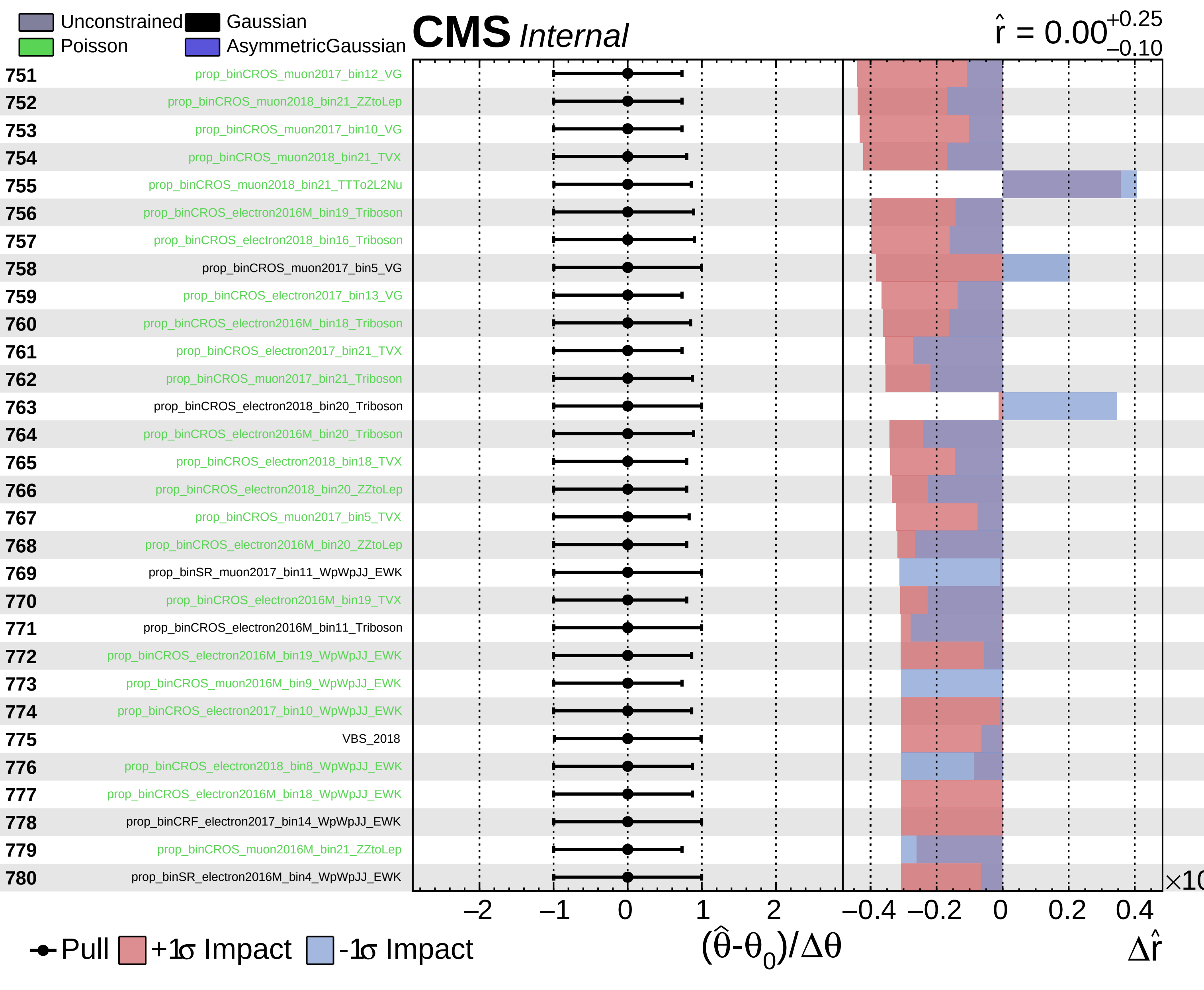


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
661 prop_binCROS_muon2018_bin20
662 prop_binCROS_electron2016M_bin9_Fake
663 prop_binCRTT_electron2016M_bin9
664 prop_binCROS_electron2017_bin21_TTTo2L2Nu
665 prop_binCROS_electron2017_bin14
666 prop_binSR_electron2017_bin20_TVX
667 prop_binCROS_muon2018_bin13_TTTo2L2Nu
668 lep_2016M
669 prop_binCRF_muon2016M_bin15_Triboson
670 prop_binCRF_muon2016M_bin15_WpWpJJ_QCD
671 prop_binCRTT_electron2016M_bin11
672 prop_binCROS_electron2016M_bin16
673 prop_binCROS_electron2018_bin16_TTTo2L2Nu
674 prop_binSR_electron2016M_bin21_WrongSign
675 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
676 prop_binCROS_muon2017_bin21_TVX
677 prop_binCROS_muon2016M_bin21_WrongSign
678 prop_binSR_electron2016M_bin4_Triboson
679 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
680 prop_binSR_electron2016M_bin21_ZZtoLep
681 prop_binSR_muon2017_bin11_TVX
682 prop_binSR_electron2017_bin20_WpWpJJ_QCD
683 prop_binCRF_electron2016M_bin11_TVX
684 prop_binCROS_electron2018_bin8_TTTo2L2Nu
685 prop_binCROS_electron2017_bin11_TVX
686 lep_2018
687 prop_binCRTT_muon2018_bin9
688 prop_binSR_muon2017_bin11_Triboson
689 prop_binCROS_muon2016M_bin18
690 prop_binCROS_muon2017_bin11_TTTo2L2Nu
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

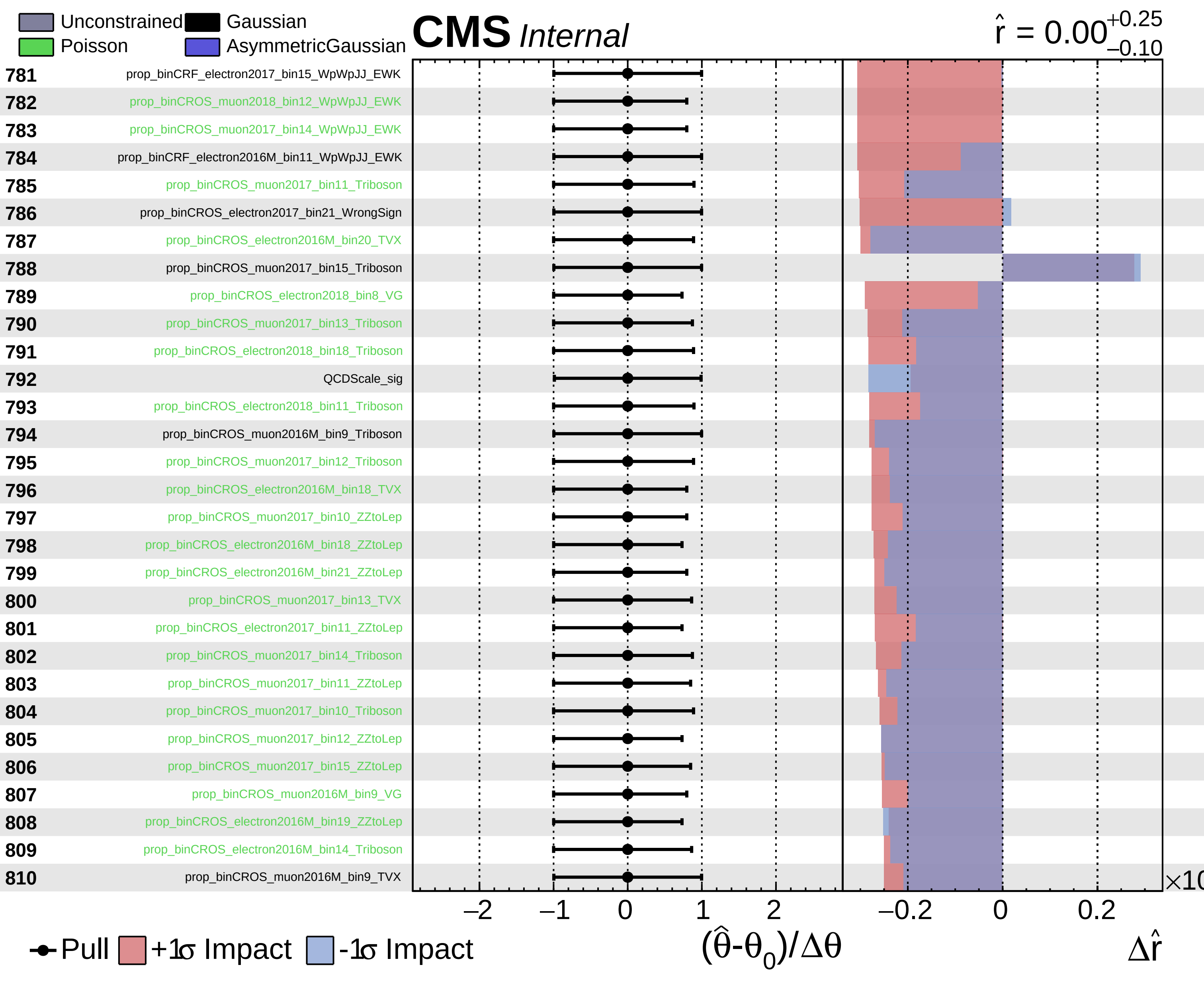


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
691 prop_binCROS_electron2016M_bin17
692 prop_binCROS_muon2017_bin12_TTTo2L2Nu
693 prop_binCRTT_muon2017_bin16
694 prop_binCRF_electron2016M_bin11_Triboson
695 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
696 prop_binCRF_electron2016M_bin14_TTTo2L2Nu
697 prop_binCRF_electron2018_bin19
698 prop_binCROS_electron2016M_bin19_TTTo2L2Nu
699 prop_binCRF_electron2017_bin19
700 prop_binCRTT_electron2017_bin5
701 prop_binCRF_electron2017_bin20
702 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
703 prop_binCROS_electron2017_bin13_WZ
704 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
705 prop_binCROS_electron2017_bin12_TTTo2L2Nu
706 prop_binCROS_muon2017_bin10_TTTo2L2Nu
707 prop_binSR_electron2016M_bin19_WrongSign
708 prop_binSR_electron2017_bin20_Triboson
709 prop_binSR_electron2016M_bin4_ZZtoLep
710 prop_binCROS_electron2017_bin11_TTTo2L2Nu
711 prop_binCROS_muon2017_bin14_TVX
712 prop_binSR_electron2017_bin21_ZZtoLep
713 prop_binCROS_electron2018_bin20_TTTo2L2Nu
714 prop_binCROS_electron2016M_bin20_WrongSign
715 prop_binCROS_electron2017_bin8_TVX
716 prop_binCROS_muon2016M_bin21_TTTo2L2Nu
717 prop_binCROS_electron2017_bin20
718 prop_binCRF_muon2016M_bin20
719 prop_binCROS_electron2016M_bin21_WrongSign
720 prop_binCROS_electron2016M_bin8_WZ
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

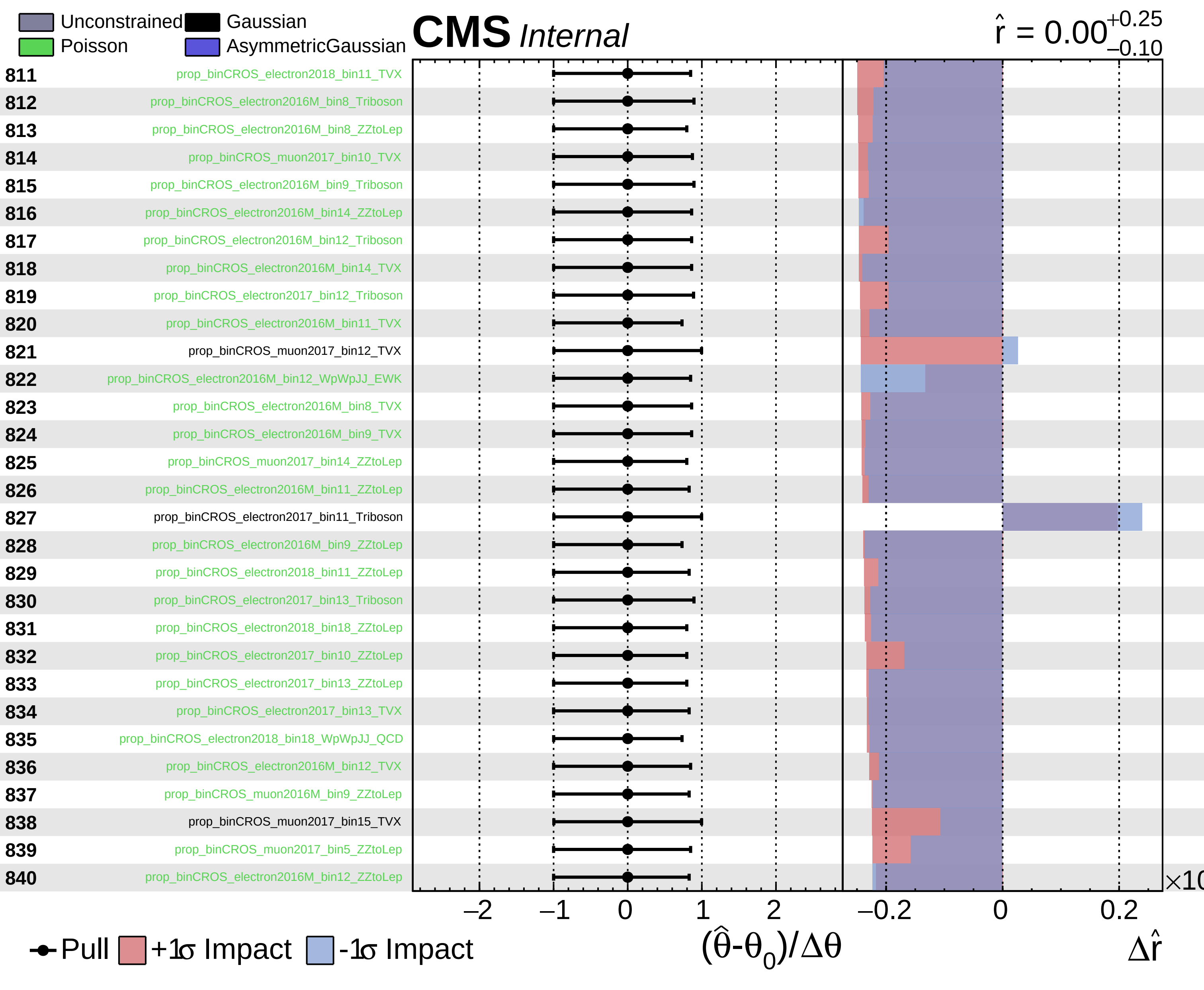

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
721 prop_binCROS_muon2018_bin21_WrongSign
722 prop_binCROS_electron2016M_bin18_TTTo2L2Nu
723 prop_binCROS_muon2018_bin16_TVX
724 prop_binCROS_electron2016M_bin7_WZ
725 prop_binCROS_muon2017_bin13_WrongSign
726 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
727 prop_binCROS_muon2017_bin21_WrongSign
728 prop_binCROS_muon2017_bin21_TTTo2L2Nu
729 prop_binCROS_electron2018_bin21_TTTo2L2Nu
730 prop_binCROS_electron2016M_bin14_VG
731 prop_binSR_electron2016M_bin4_TVX
732 prop_binSR_electron2016M_bin4_WpWpJJ_QCD
733 prop_binCRF_electron2016M_bin12_ZZtoLep
734 prop_binCROS_electron2018_bin11_TTTo2L2Nu
735 prop_binSR_electron2017_bin20_ZZtoLep
736 prop_binCROS_electron2017_bin21_Triboson
737 prop_binCRF_electron2017_bin15_Triboson
738 prop_binCROS_electron2016M_bin21_Triboson
739 prop_binCROS_electron2018_bin21_Triboson
740 prop_binCROS_muon2018_bin21_Triboson
741 FES
742 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
743 prop_binCRF_electron2016M_bin14_TVX
744 prop_binCROS_muon2017_bin11_VG
745 prop_binCROS_electron2018_bin16_ZZtoLep
746 prop_binCROS_electron2018_bin16_TVX
747 prop_binCROS_muon2017_bin11_TVX
748 prop_binCRF_electron2016M_bin14_Triboson
749 prop_binCROS_muon2016M_bin21_Triboson
750 prop_binSR_muon2017_bin11_ZZtoLep
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



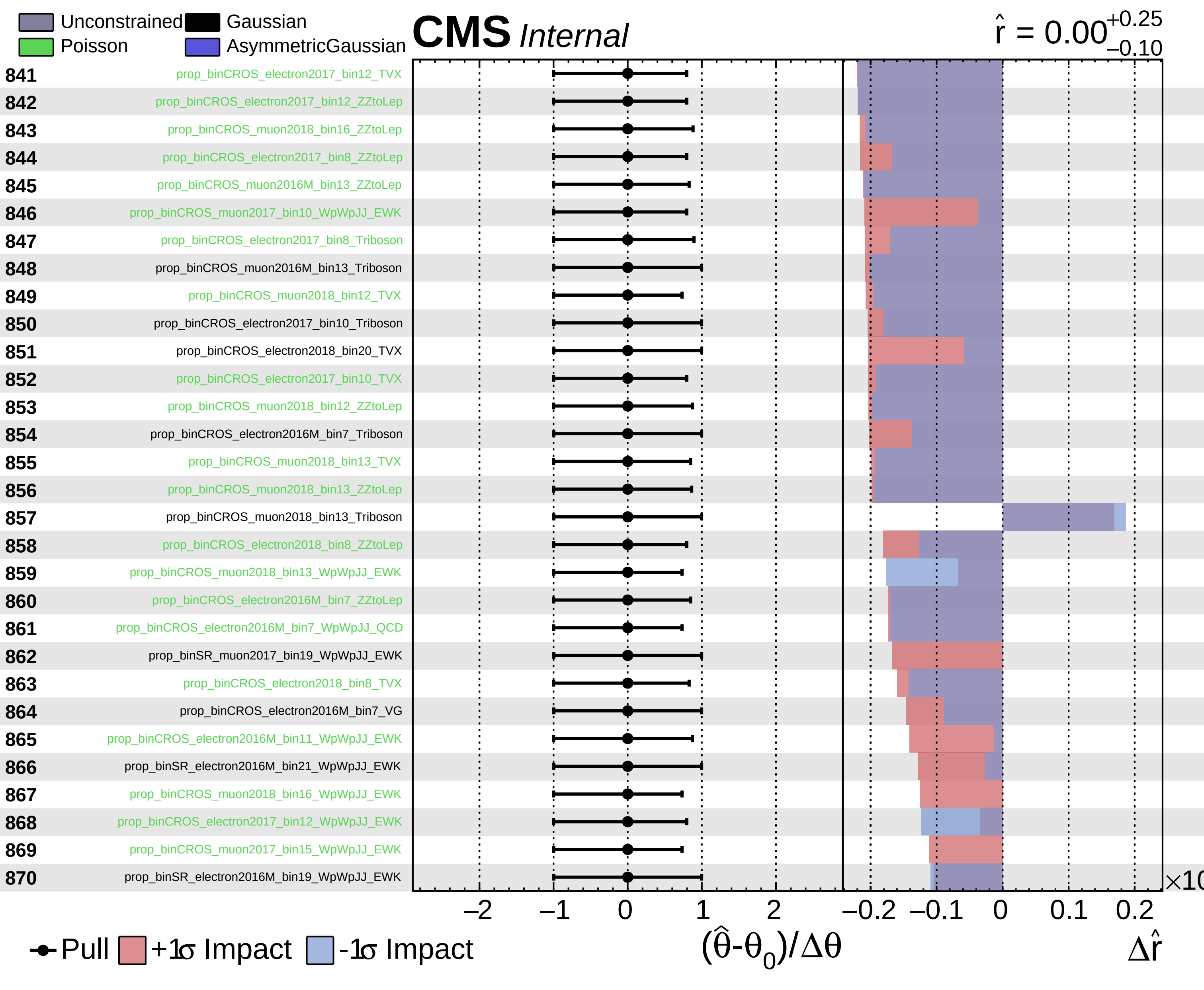
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.25 −0.10
751 prop_binCROS_muon2017_bin12_VG
752 prop_binCROS_muon2018_bin21_ZZtoLep
753 prop_binCROS_muon2017_bin10_VG
754 prop_binCROS_muon2018_bin21_TVX
755 prop_binCROS_muon2018_bin21_TTTo2L2Nu
756 prop_binCROS_electron2016M_bin19_Triboson
757 prop_binCROS_electron2018_bin16_Triboson
758 prop_binCROS_muon2017_bin5_VG
759 prop_binCROS_electron2017_bin13_VG
760 prop_binCROS_electron2016M_bin18_Triboson
761 prop_binCROS_electron2017_bin21_TVX
762 prop_binCROS_muon2017_bin21_Triboson
763 prop_binCROS_electron2018_bin20_Triboson
764 prop_binCROS_electron2016M_bin20_Triboson
765 prop_binCROS_electron2018_bin18_TVX
766 prop_binCROS_electron2018_bin20_ZZtoLep
767 prop_binCROS_muon2017_bin5_TVX
768 prop_binCROS_electron2016M_bin20_ZZtoLep
769 prop_binSR_muon2017_bin11_WpWpJJ_EWK
770 prop_binCROS_electron2016M_bin19_TVX
771 prop_binCROS_electron2016M_bin11_Triboson
772 prop_binCROS_electron2016M_bin19_WpWpJJ_EWK
773 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
774 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
775 VBS_2018
776 prop_binCROS_electron2018_bin8_WpWpJJ_EWK
777 prop_binCROS_electron2016M_bin18_WpWpJJ_EWK
778 prop_binCRF_electron2017_bin14_WpWpJJ_EWK
779 prop_binCROS_muon2016M_bin21_ZZtoLep
780 prop_binSR_electron2016M_bin4_WpWpJJ_EWK
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



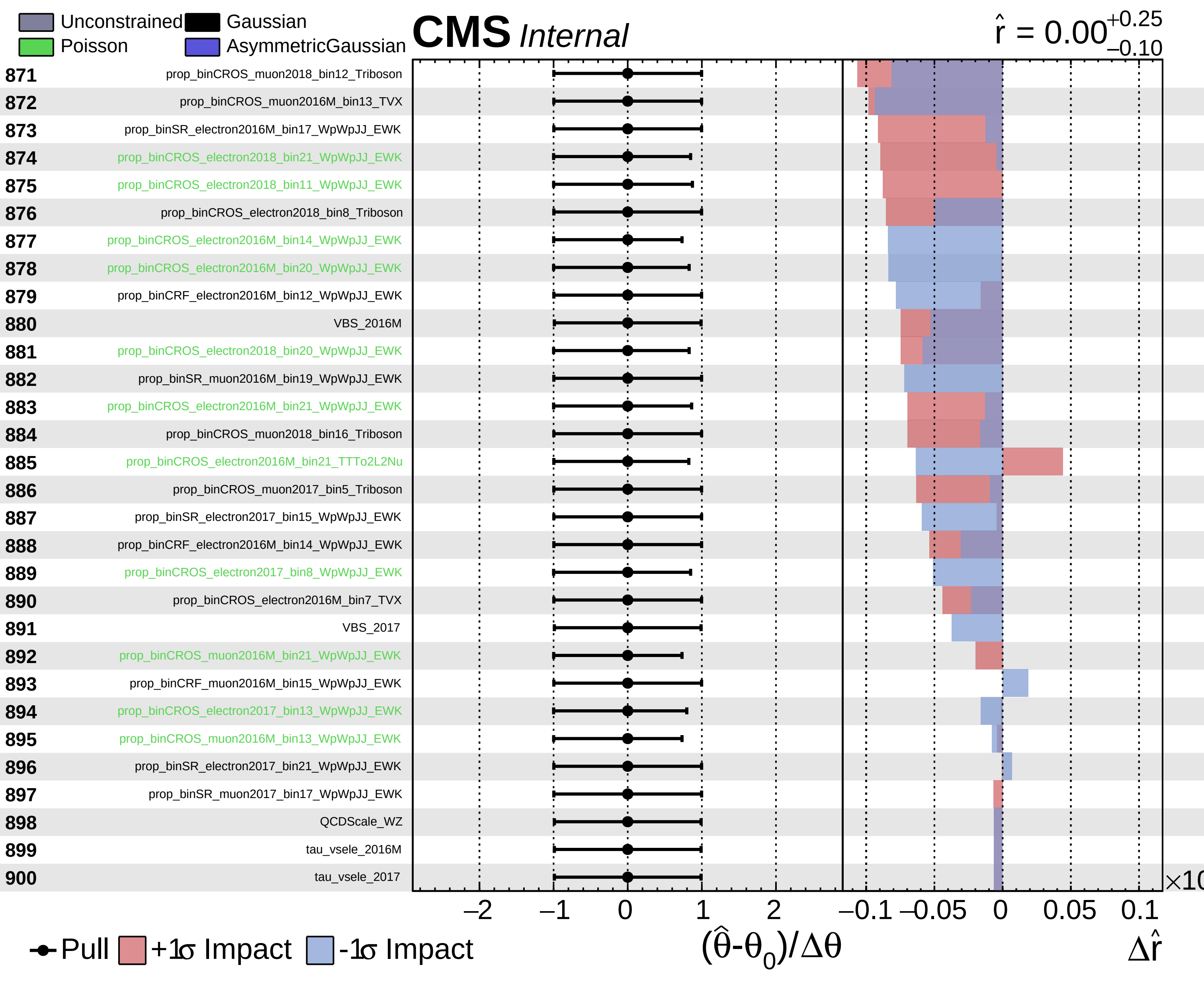
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
781 prop_binCRF_electron2017_bin15_WpWpJJ_EWK
782 prop_binCROS_muon2018_bin12_WpWpJJ_EWK
783 prop_binCROS_muon2017_bin14_WpWpJJ_EWK
784 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
785 prop_binCROS_muon2017_bin11_Triboson
786 prop_binCROS_electron2017_bin21_WrongSign
787 prop_binCROS_electron2016M_bin20_TVX
788 prop_binCROS_muon2017_bin15_Triboson
789 prop_binCROS_electron2018_bin8_VG
790 prop_binCROS_muon2017_bin13_Triboson
791 prop_binCROS_electron2018_bin18_Triboson
792 QCDScale_sig
793 prop_binCROS_electron2018_bin11_Triboson
794 prop_binCROS_muon2016M_bin9_Triboson
795 prop_binCROS_muon2017_bin12_Triboson
796 prop_binCROS_electron2016M_bin18_TVX
797 prop_binCROS_muon2017_bin10_ZZtoLep
798 prop_binCROS_electron2016M_bin18_ZZtoLep
799 prop_binCROS_electron2016M_bin21_ZZtoLep
800 prop_binCROS_muon2017_bin13_TVX
801 prop_binCROS_electron2017_bin11_ZZtoLep
802 prop_binCROS_muon2017_bin14_Triboson
803 prop_binCROS_muon2017_bin11_ZZtoLep
804 prop_binCROS_muon2017_bin10_Triboson
805 prop_binCROS_muon2017_bin12_ZZtoLep
806 prop_binCROS_muon2017_bin15_ZZtoLep
807 prop_binCROS_muon2016M_bin9_VG
808 prop_binCROS_electron2016M_bin19_ZZtoLep
809 prop_binCROS_electron2016M_bin14_Triboson
810 prop_binCROS_muon2016M_bin9_TVX
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



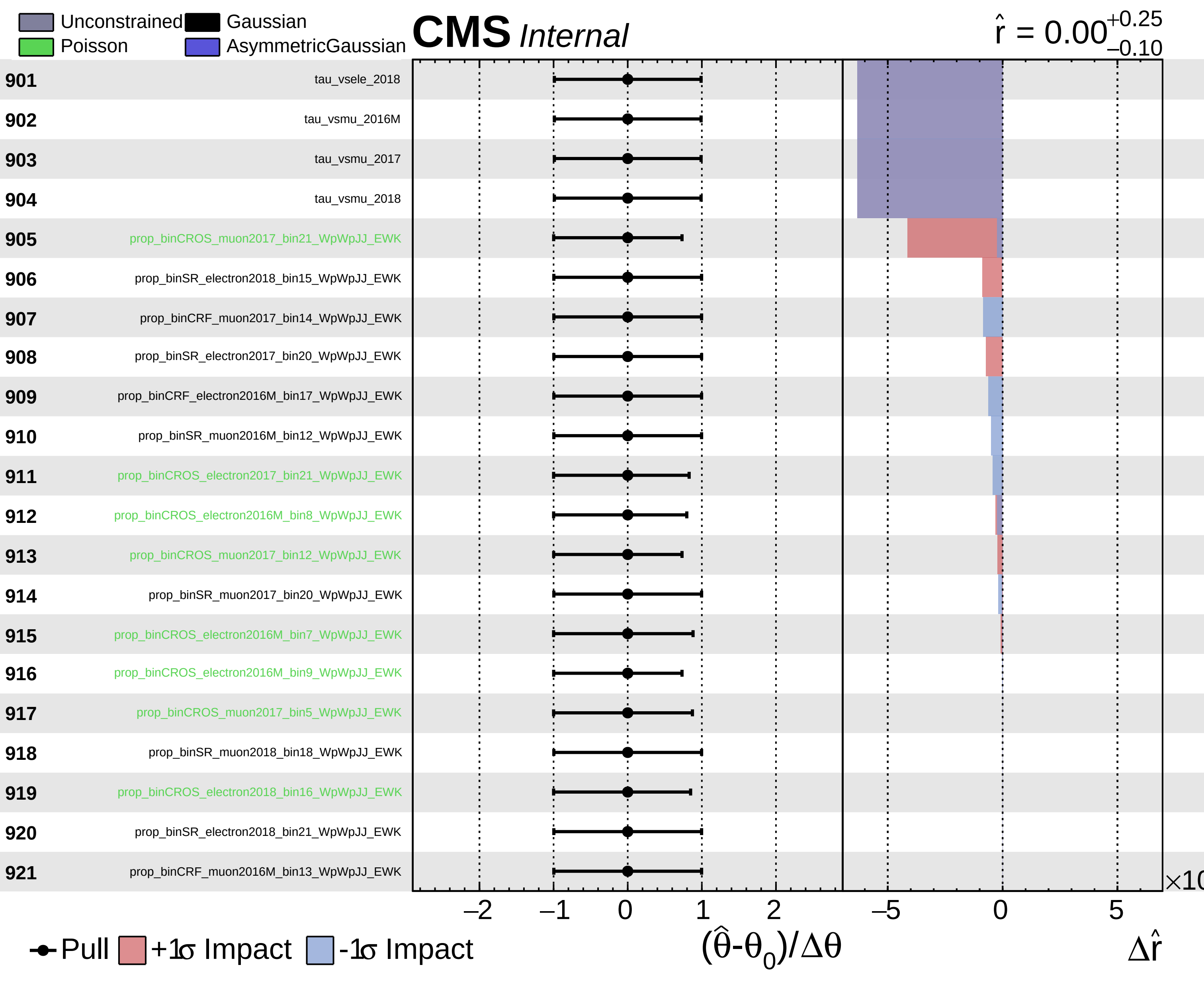
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
811 prop_binCROS_electron2018_bin11_TVX
812 prop_binCROS_electron2016M_bin8_Triboson
813 prop_binCROS_electron2016M_bin8_ZZtoLep
814 prop_binCROS_muon2017_bin10_TVX
815 prop_binCROS_electron2016M_bin9_Triboson
816 prop_binCROS_electron2016M_bin14_ZZtoLep
817 prop_binCROS_electron2016M_bin12_Triboson
818 prop_binCROS_electron2016M_bin14_TVX
819 prop_binCROS_electron2017_bin12_Triboson
820 prop_binCROS_electron2016M_bin11_TVX
821 prop_binCROS_muon2017_bin12_TVX
822 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
823 prop_binCROS_electron2016M_bin8_TVX
824 prop_binCROS_electron2016M_bin9_TVX
825 prop_binCROS_muon2017_bin14_ZZtoLep
826 prop_binCROS_electron2016M_bin11_ZZtoLep
827 prop_binCROS_electron2017_bin11_Triboson
828 prop_binCROS_electron2016M_bin9_ZZtoLep
829 prop_binCROS_electron2018_bin11_ZZtoLep
830 prop_binCROS_electron2017_bin13_Triboson
831 prop_binCROS_electron2018_bin18_ZZtoLep
832 prop_binCROS_electron2017_bin10_ZZtoLep
833 prop_binCROS_electron2017_bin13_ZZtoLep
834 prop_binCROS_electron2017_bin13_TVX
835 prop_binCROS_electron2018_bin18_WpWpJJ_QCD
836 prop_binCROS_electron2016M_bin12_TVX
837 prop_binCROS_muon2016M_bin9_ZZtoLep
838 prop_binCROS_muon2017_bin15_TVX
839 prop_binCROS_muon2017_bin5_ZZtoLep
840 prop_binCROS_electron2016M_bin12_ZZtoLep
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
841 prop_binCROS_electron2017_bin12_TVX
842 prop_binCROS_electron2017_bin12_ZZtoLep
843 prop_binCROS_muon2018_bin16_ZZtoLep
844 prop_binCROS_electron2017_bin8_ZZtoLep
845 prop_binCROS_muon2016M_bin13_ZZtoLep
846 prop_binCROS_muon2017_bin10_WpWpJJ_EWK
847 prop_binCROS_electron2017_bin8_Triboson
848 prop_binCROS_muon2016M_bin13_Triboson
849 prop_binCROS_muon2018_bin12_TVX
850 prop_binCROS_electron2017_bin10_Triboson
851 prop_binCROS_electron2018_bin20_TVX
852 prop_binCROS_electron2017_bin10_TVX
853 prop_binCROS_muon2018_bin12_ZZtoLep
854 prop_binCROS_electron2016M_bin7_Triboson
855 prop_binCROS_muon2018_bin13_TVX
856 prop_binCROS_muon2018_bin13_ZZtoLep
857 prop_binCROS_muon2018_bin13_Triboson
858 prop_binCROS_electron2018_bin8_ZZtoLep
859 prop_binCROS_muon2018_bin13_WpWpJJ_EWK
860 prop_binCROS_electron2016M_bin7_ZZtoLep
861 prop_binCROS_electron2016M_bin7_WpWpJJ_QCD
862 prop_binSR_muon2017_bin19_WpWpJJ_EWK
863 prop_binCROS_electron2018_bin8_TVX
864 prop_binCROS_electron2016M_bin7_VG
865 prop_binCROS_electron2016M_bin11_WpWpJJ_EWK
866 prop_binSR_electron2016M_bin21_WpWpJJ_EWK
867 prop_binCROS_muon2018_bin16_WpWpJJ_EWK
868 prop_binCROS_electron2017_bin12_WpWpJJ_EWK
869 prop_binCROS_muon2017_bin15_WpWpJJ_EWK
870 prop_binSR_electron2016M_bin19_WpWpJJ_EWK
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
871 prop_binCROS_muon2018_bin12_Triboson
872 prop_binCROS_muon2016M_bin13_TVX
873 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
874 prop_binCROS_electron2018_bin21_WpWpJJ_EWK
875 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
876 prop_binCROS_electron2018_bin8_Triboson
877 prop_binCROS_electron2016M_bin14_WpWpJJ_EWK
878 prop_binCROS_electron2016M_bin20_WpWpJJ_EWK
879 prop_binCRF_electron2016M_bin12_WpWpJJ_EWK
880 VBS_2016M
881 prop_binCROS_electron2018_bin20_WpWpJJ_EWK
882 prop_binSR_muon2016M_bin19_WpWpJJ_EWK
883 prop_binCROS_electron2016M_bin21_WpWpJJ_EWK
884 prop_binCROS_muon2018_bin16_Triboson
885 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
886 prop_binCROS_muon2017_bin5_Triboson
887 prop_binSR_electron2017_bin15_WpWpJJ_EWK
888 prop_binCRF_electron2016M_bin14_WpWpJJ_EWK
889 prop_binCROS_electron2017_bin8_WpWpJJ_EWK
890 prop_binCROS_electron2016M_bin7_TVX
891 VBS_2017
892 prop_binCROS_muon2016M_bin21_WpWpJJ_EWK
893 prop_binCRF_muon2016M_bin15_WpWpJJ_EWK
894 prop_binCROS_electron2017_bin13_WpWpJJ_EWK
895 prop_binCROS_muon2016M_bin13_WpWpJJ_EWK
896 prop_binSR_electron2017_bin21_WpWpJJ_EWK
897 prop_binSR_muon2017_bin17_WpWpJJ_EWK
898 QCDScale_WZ
899 tau_vsele_2016M
900 tau_vsele_2017
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.25}_{-0.10}$
901 tau_vsele_2018
902 tau_vsmu_2016M
903 tau_vsmu_2017
904 tau_vsmu_2018
905 prop_binCROS_muon2017_bin21_WpWpJJ_EWK
906 prop_binSR_electron2018_bin15_WpWpJJ_EWK
907 prop_binCRF_muon2017_bin14_WpWpJJ_EWK
908 prop_binSR_electron2017_bin20_WpWpJJ_EWK
909 prop_binCRF_electron2016M_bin17_WpWpJJ_EWK
910 prop_binSR_muon2016M_bin12_WpWpJJ_EWK
911 prop_binCROS_electron2017_bin21_WpWpJJ_EWK
912 prop_binCROS_electron2016M_bin8_WpWpJJ_EWK
913 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
914 prop_binSR_muon2017_bin20_WpWpJJ_EWK
915 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
916 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
917 prop_binCROS_muon2017_bin5_WpWpJJ_EWK
918 prop_binSR_muon2018_bin18_WpWpJJ_EWK
919 prop_binCROS_electron2018_bin16_WpWpJJ_EWK
920 prop_binSR_electron2018_bin21_WpWpJJ_EWK
921 prop_binCRF_muon2016M_bin13_WpWpJJ_EWK
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$